

DTU





Elite Summer School Robotics and Entrepreneurship
August 14, 2025
Odense, Denmark

Lazaros Nalpantidis

Perceiving Depth from Images

Stereoscopic Vision and the Stereo Correspondence Problem

some slides borrowed or adapted from:

- Noah Snavely
- Aaron Bobick
- Antonio Torralba



Lazaros Nalpantidis

- **Professor** of Autonomous Systems
- **Head of Group** of Perception & Cognition for Autonomous Systems
- ~~Head of Studies~~ of MSc in Autonomous Systems





Autonomous Robots in Challenging Environments e.g. Agriculture, Construction

- **Autonomous Robotic Systems** lie at the intersection of:
 - Robotics,
 - Machine Learning (AI),
 - Sensing & Perception.

- An autonomous system can:
 - **perceive** its environment,
 - operate **robustly** without human control over long periods of time,
 - learn to **develop & adapt** in un-foreseen or changing situations.

- DL for Situational Awareness: Training Data is never enough ☹️
 - Synthetic Data generation
 - Self-Supervised Learning
 - Multi-task Learning
 - Life-long Learning



- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary



Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary



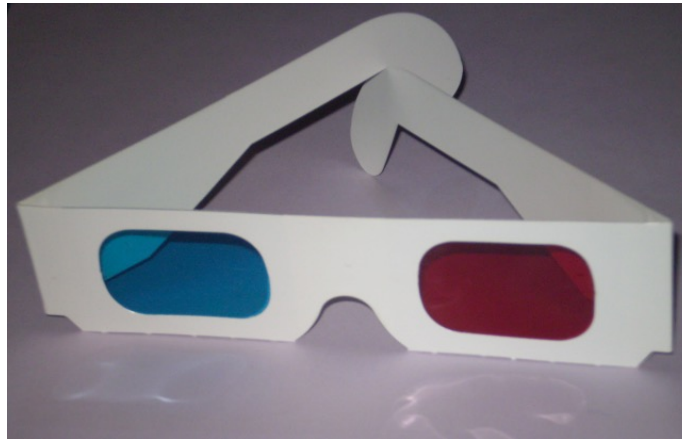
What is Stereo Vision?





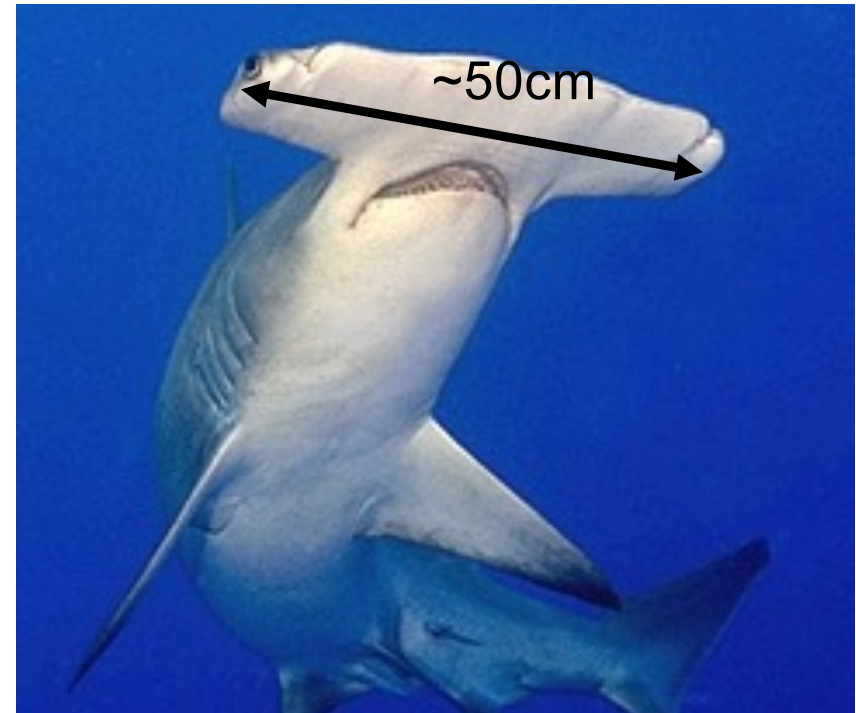
What is Stereo Vision?

- 3D cinema
- 3D television
- ...





What is Stereo Vision?

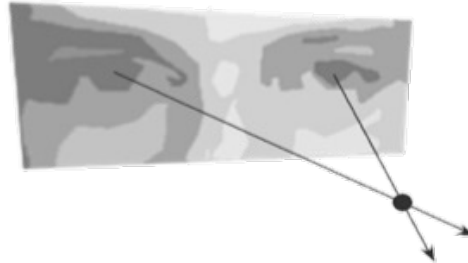


What is Stereo Vision?



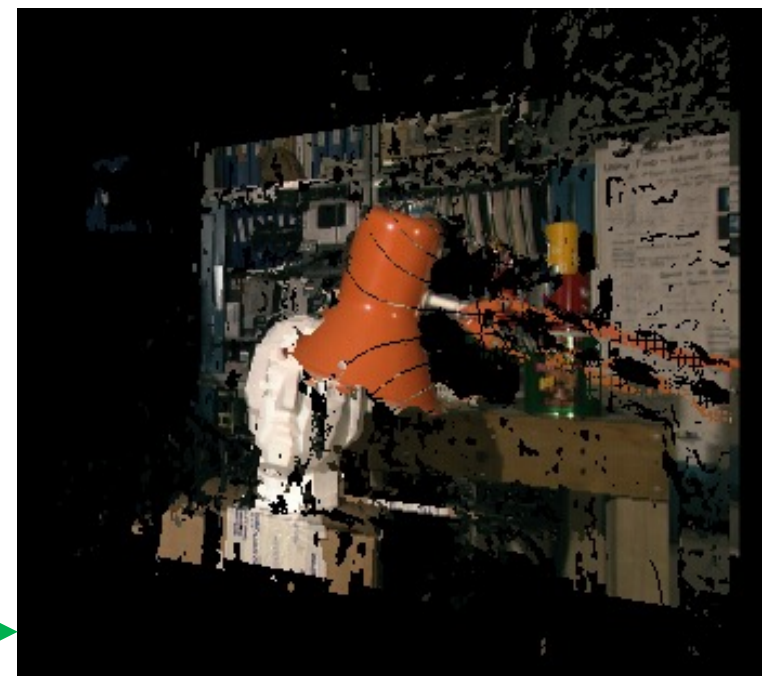
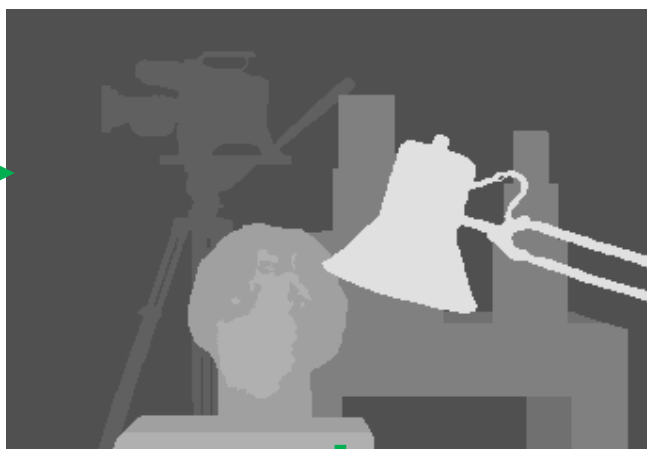
“... the mind perceives an object of three-dimensions by means of the two dissimilar pictures projected by it on the two retinae...”

Sir Charles Wheatstone, 1838





Stereo Vision Computation



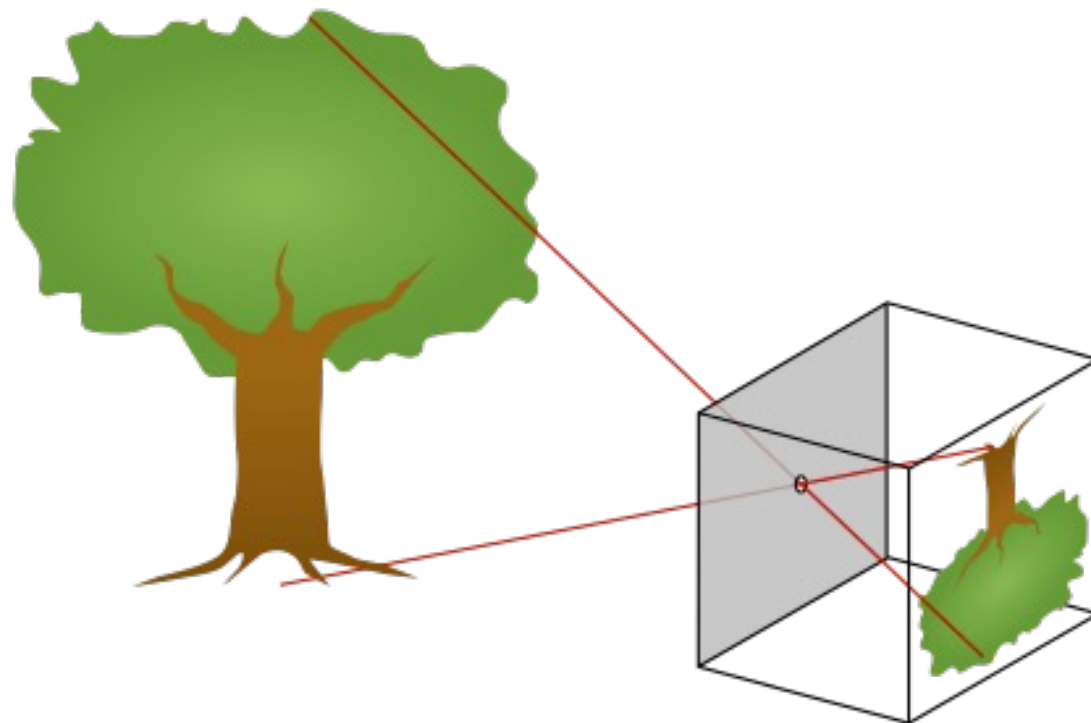


Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary

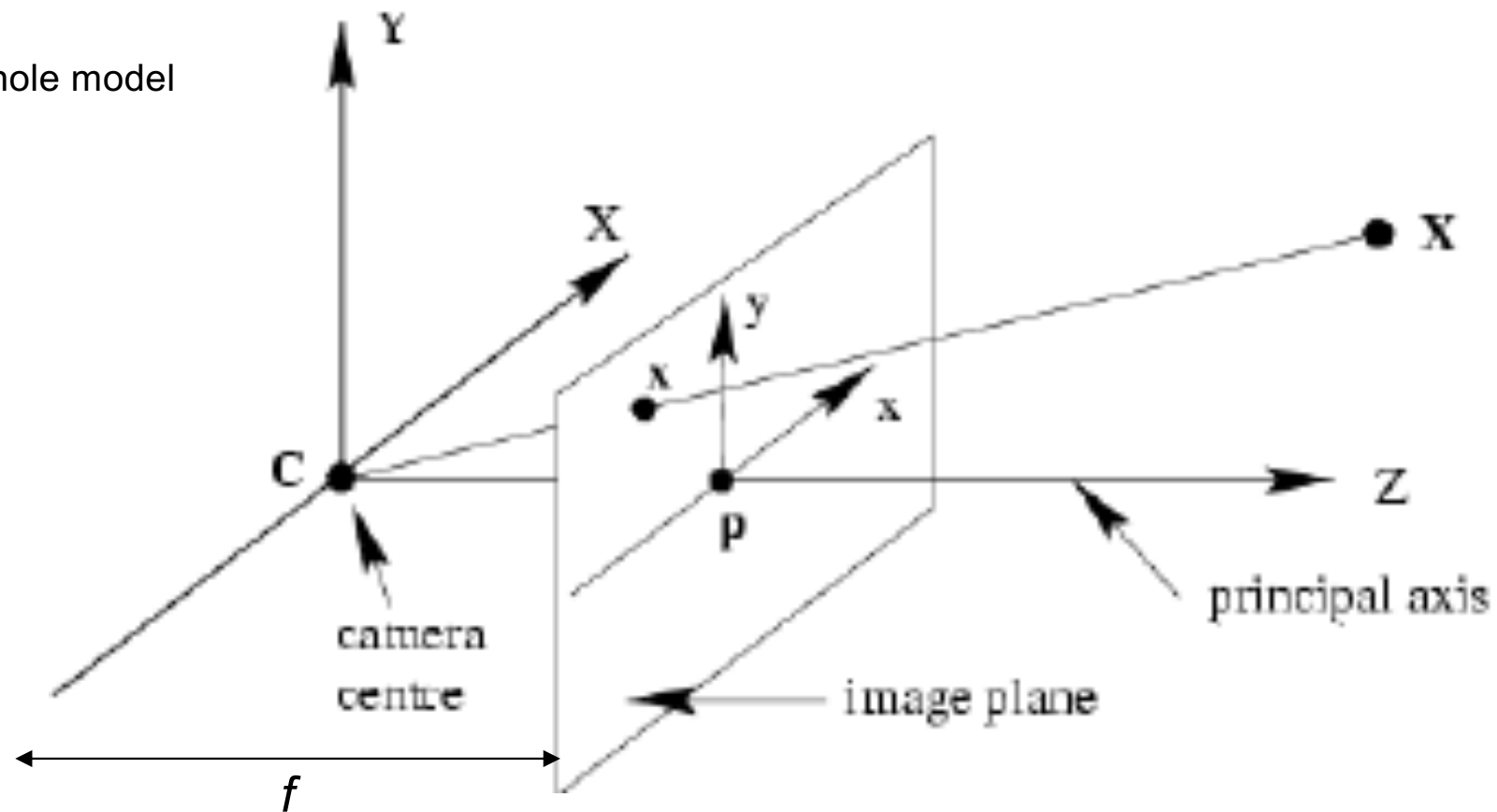
Camera Model

- Pinhole model

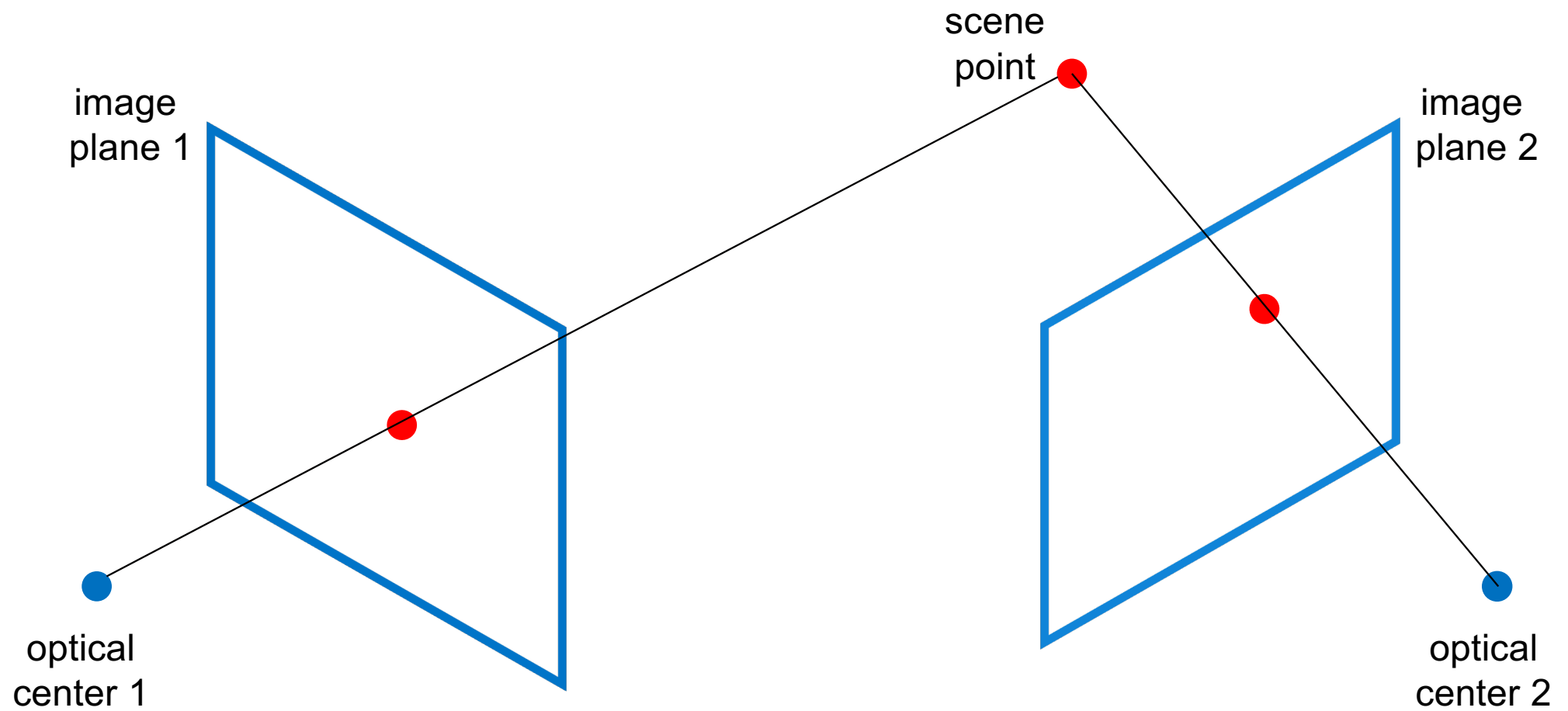


Camera Model

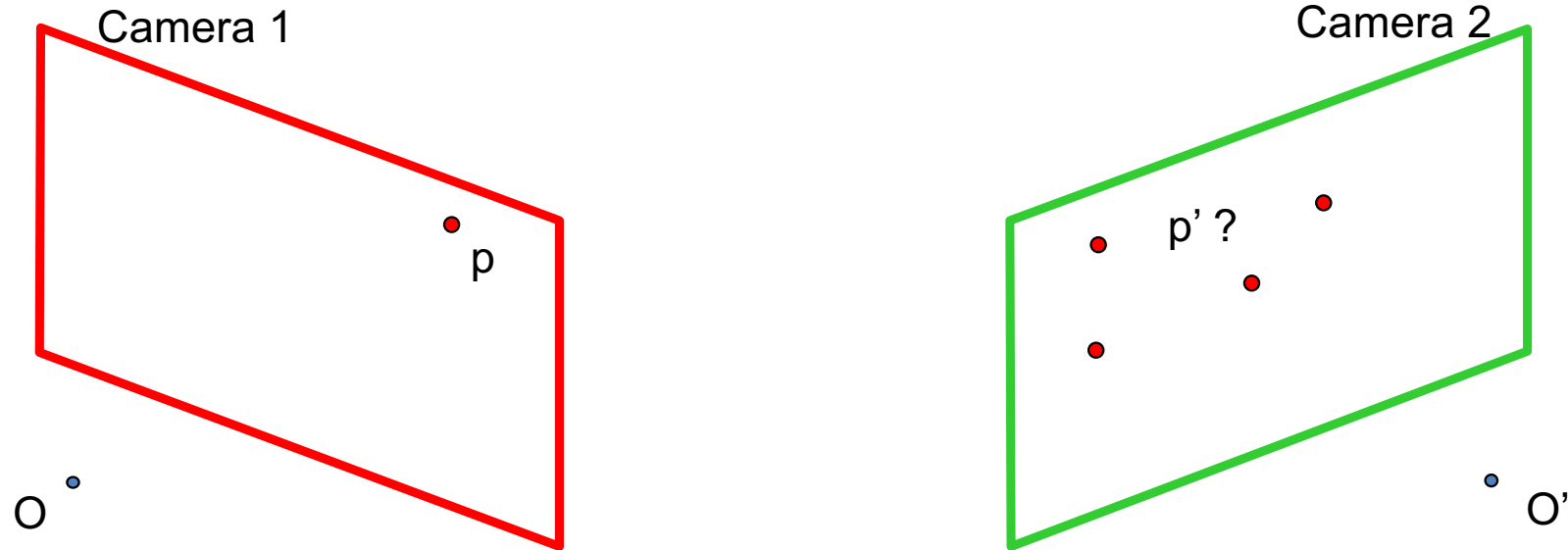
- Pinhole model



Epipolar Geometry

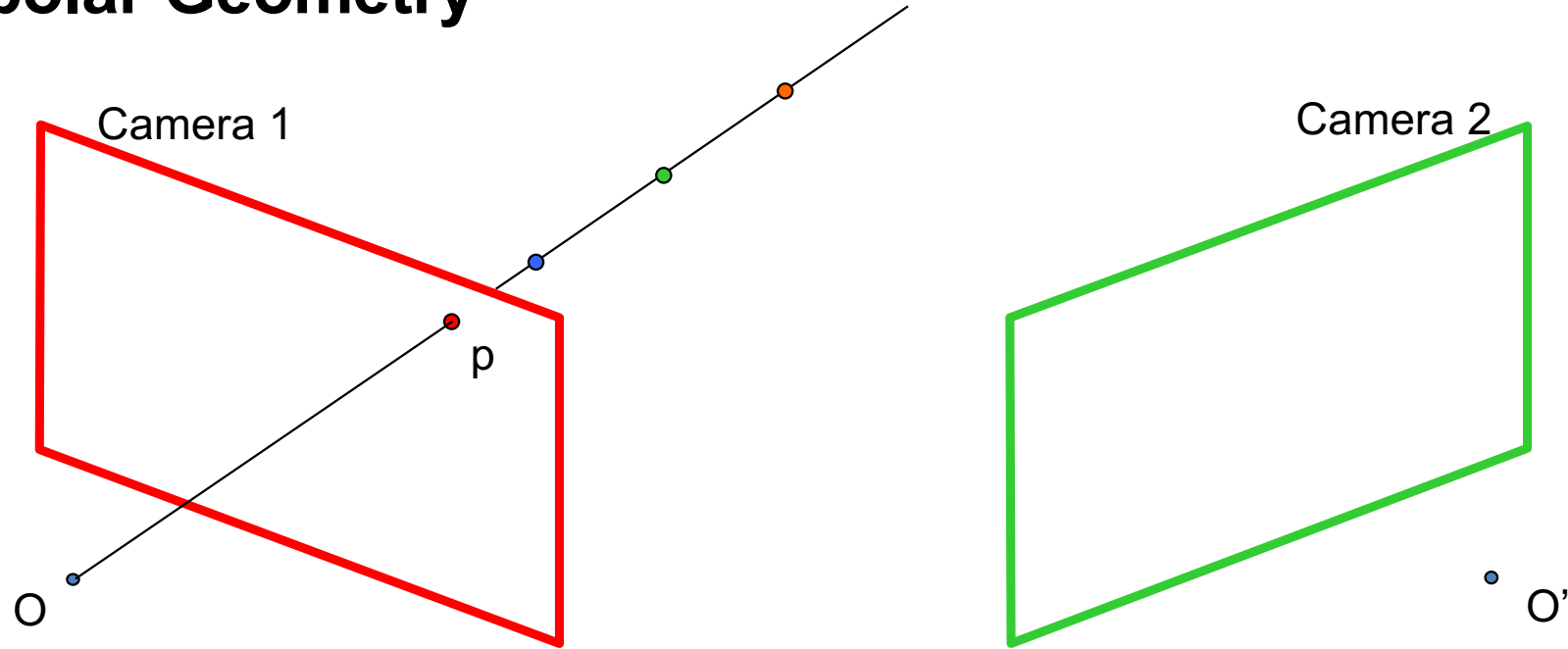


Epipolar Geometry

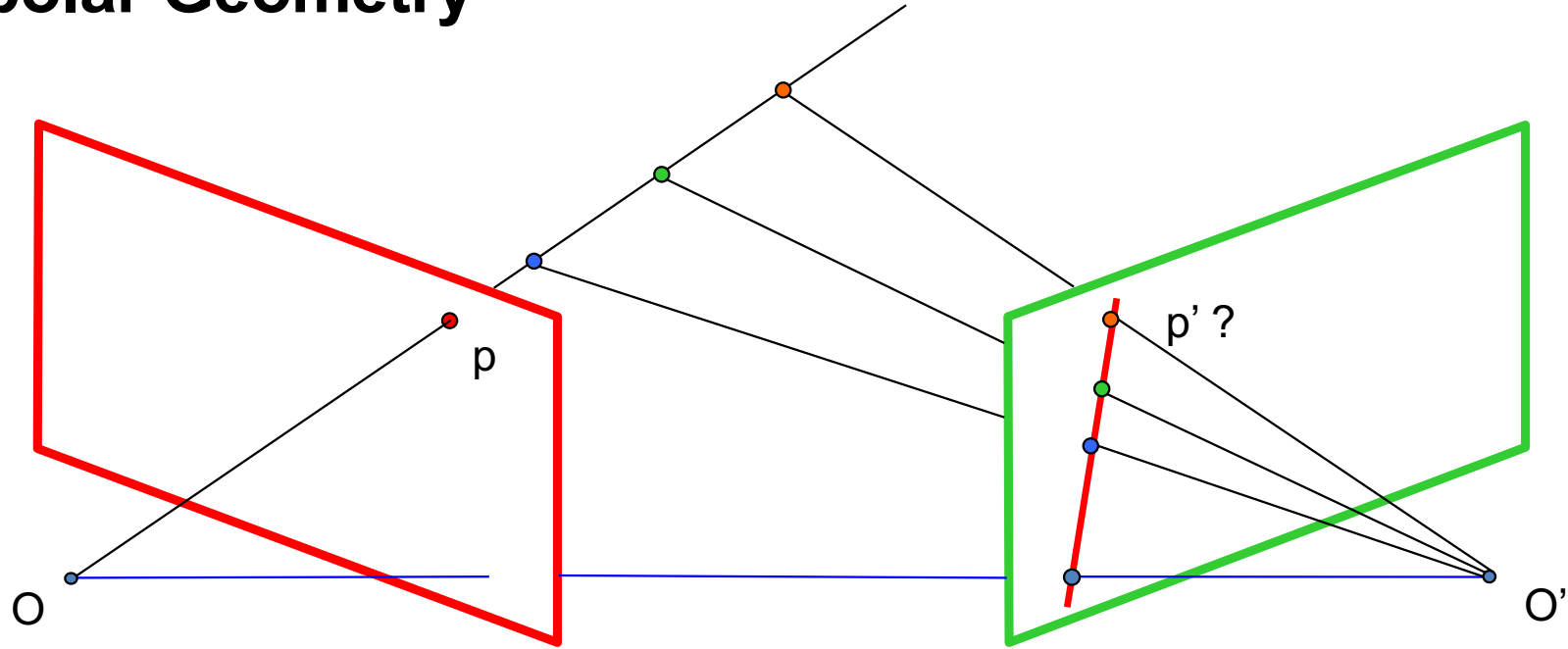


If we see a point in camera 1, are there any constraints on where we will find it on camera 2?

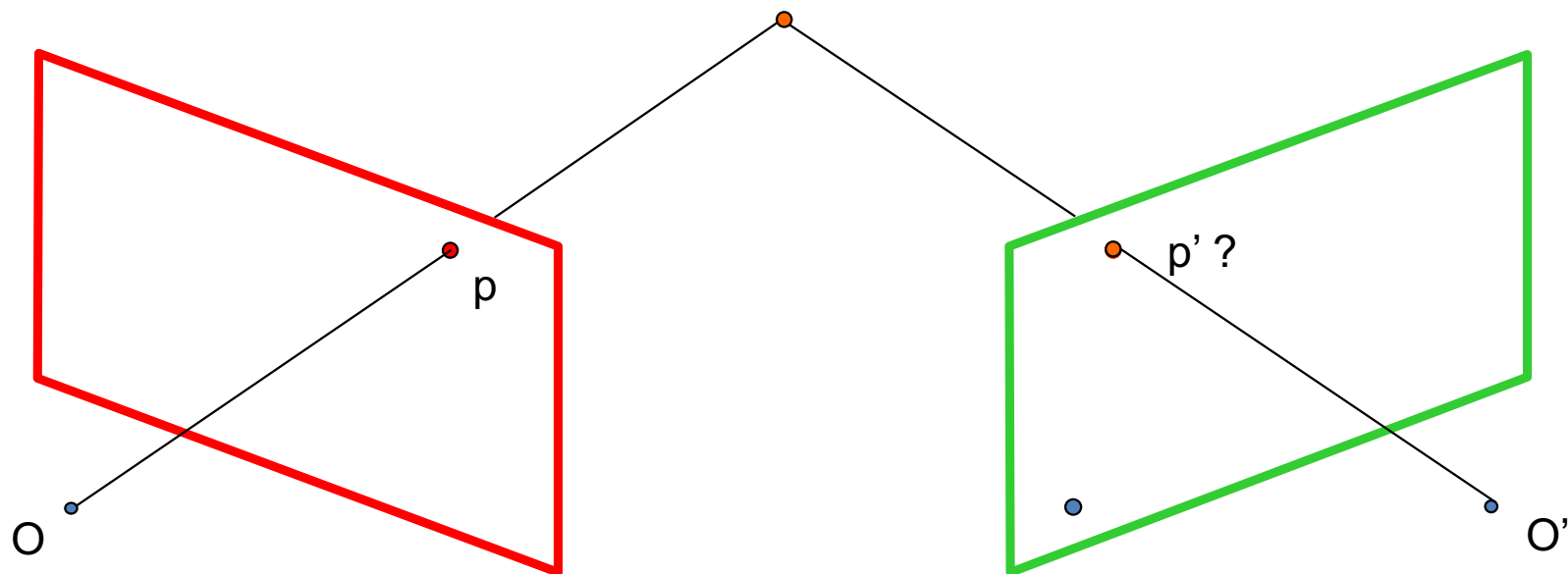
Epipolar Geometry



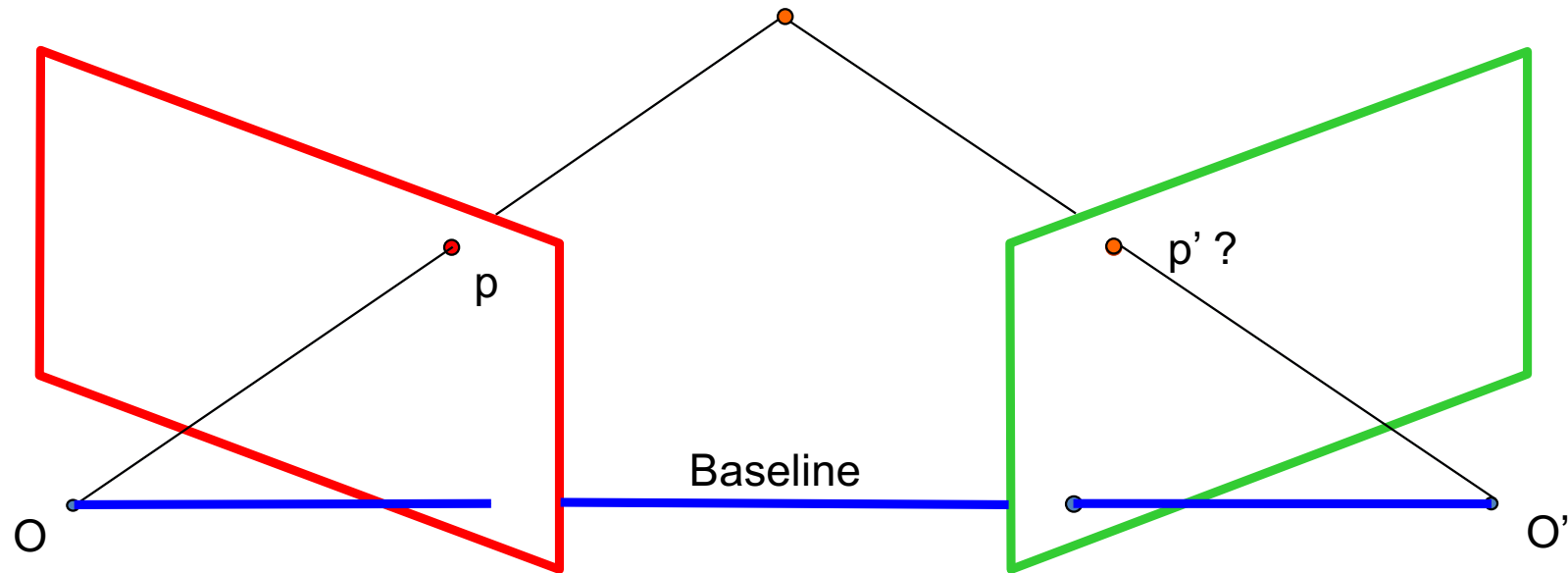
If we see a point in camera 1, are there any constraints on where we will find it on camera 2?



Epipolar Geometry

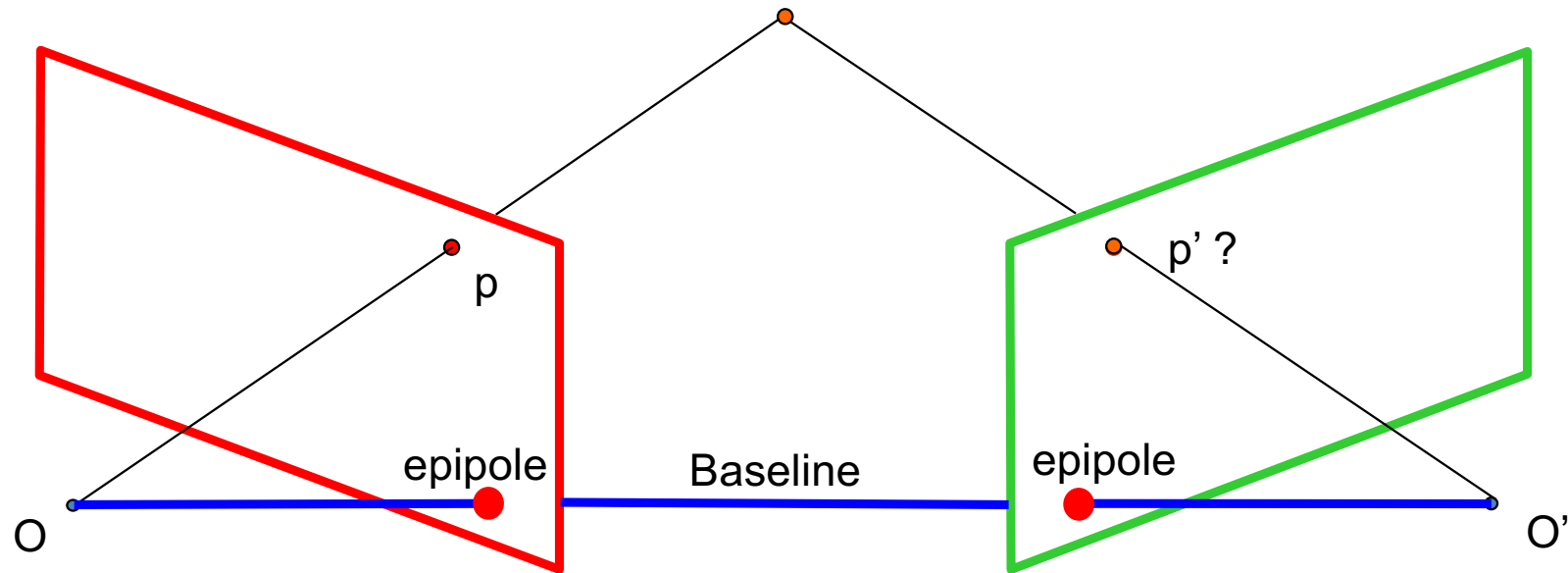


Epipolar Geometry



Baseline: the line connecting the two camera centers

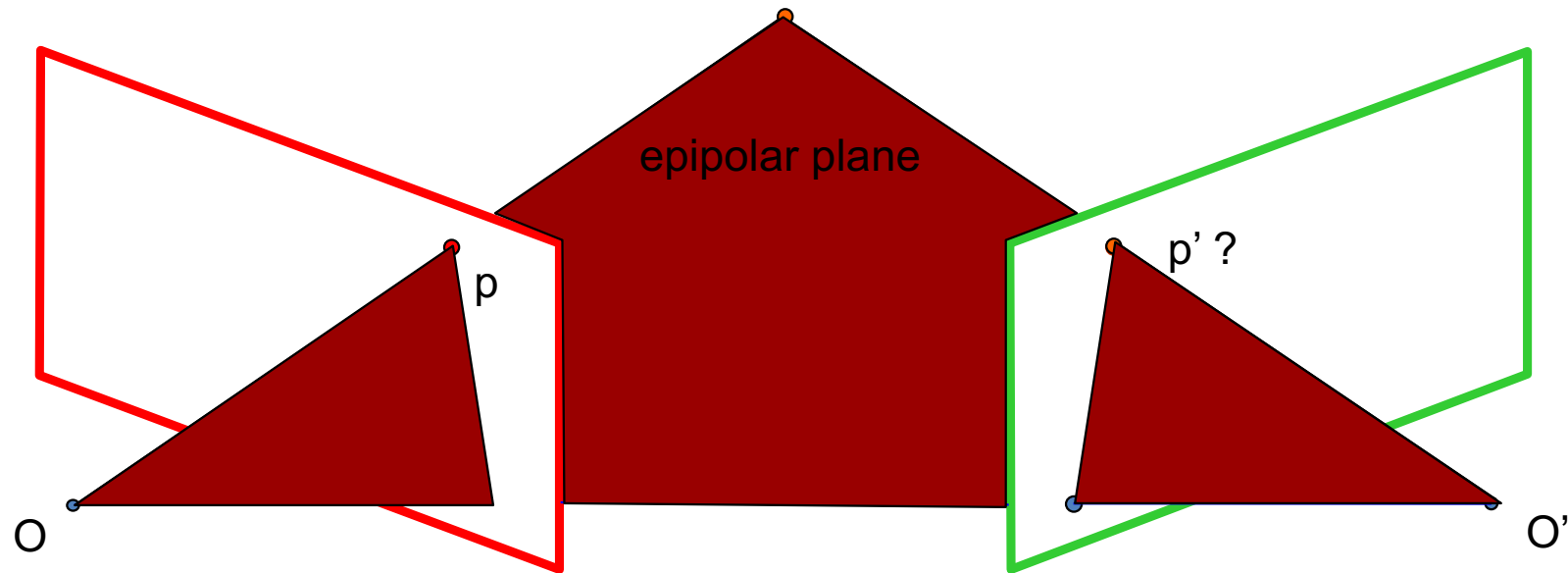
Epipolar Geometry



Baseline: the line connecting the two camera centers

Epipole: point of intersection of *baseline* with the image plane

Epipolar Geometry

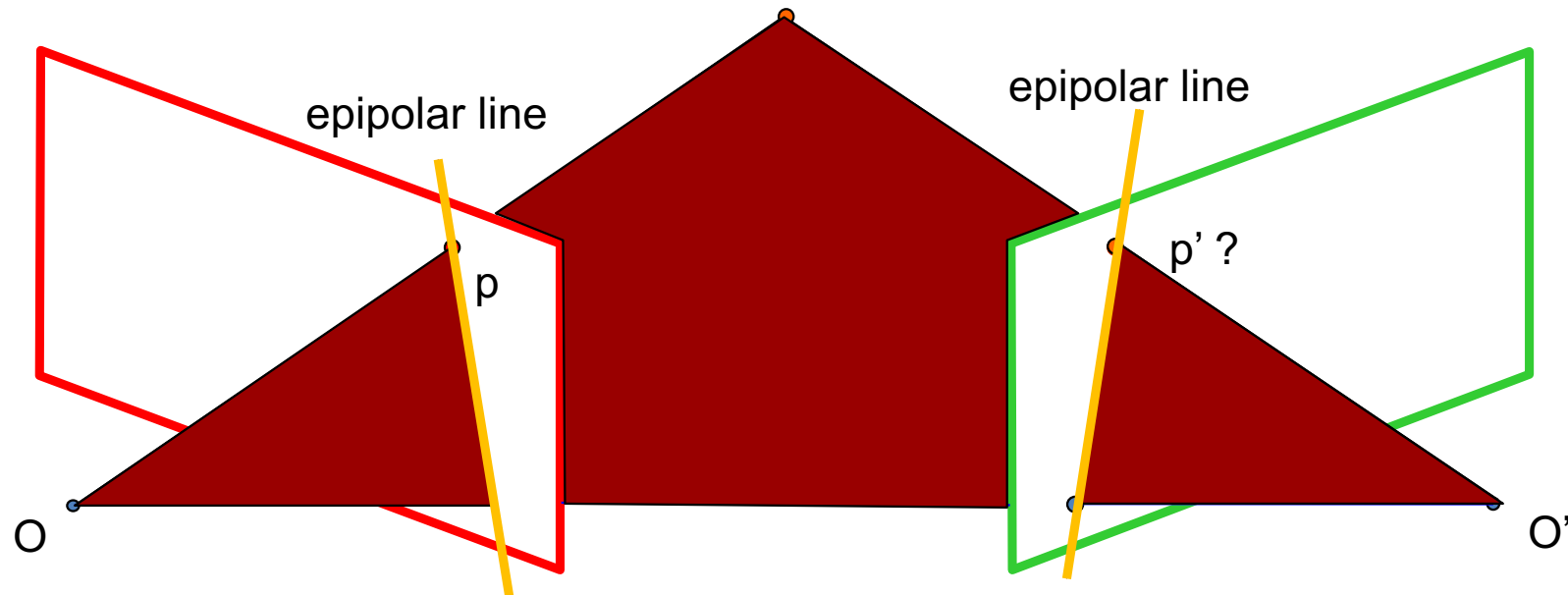


Baseline: the line connecting the two camera centers

Epipole: point of intersection of *baseline* with the image plane

Epipolar plane: the plane that contains the two camera centers and a 3D point in the world

Epipolar Geometry



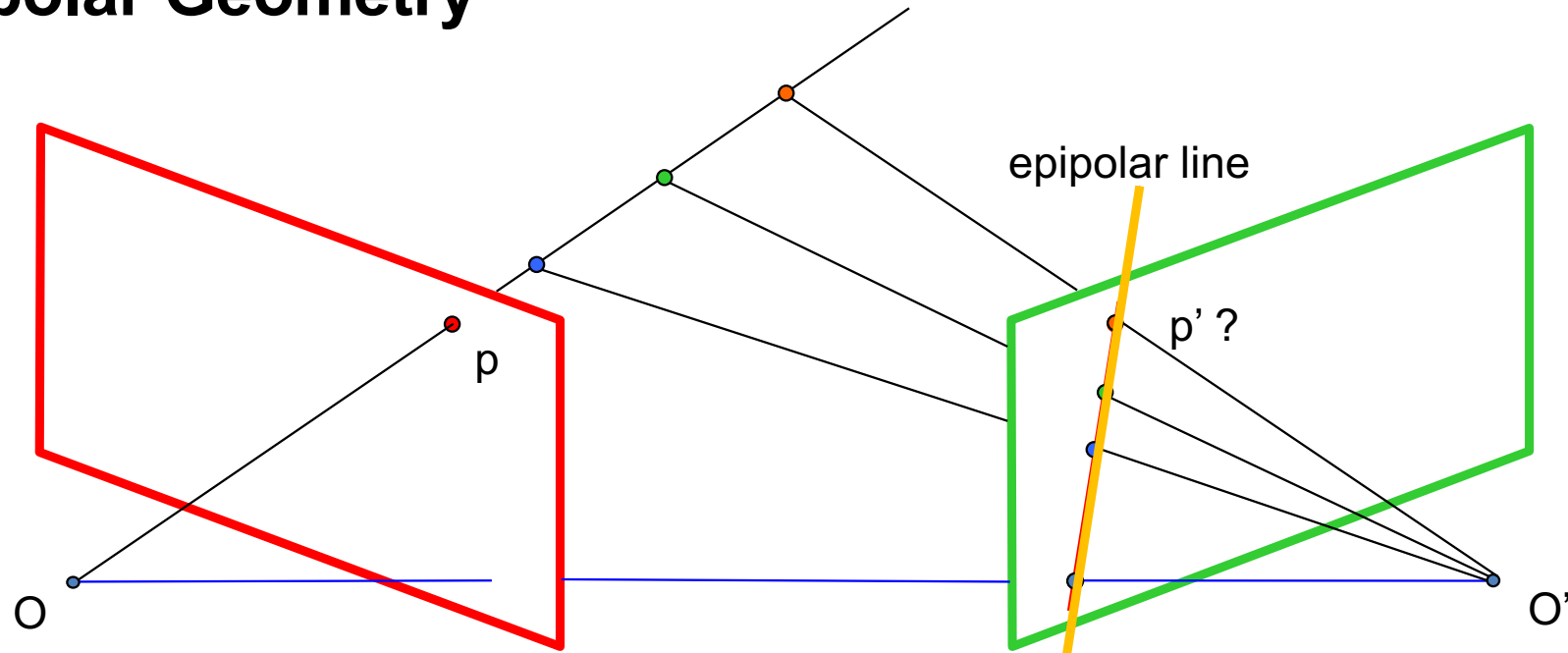
Baseline: the line connecting the two camera centers

Epipole: point of intersection of *baseline* with the image plane

Epipolar plane: the plane that contains the two camera centers and a 3D point in the world

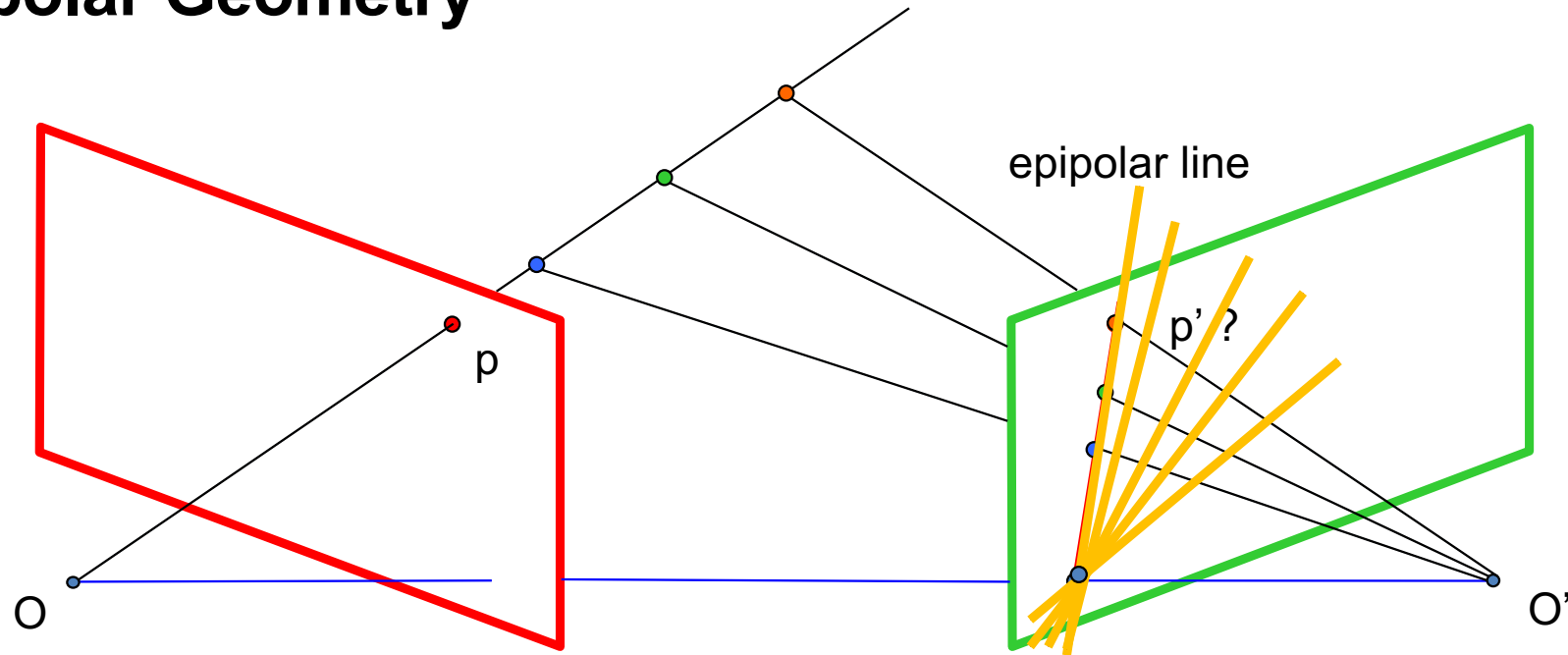
Epipolar line: intersection of the *epipolar plane* with each image plane

Epipolar Geometry



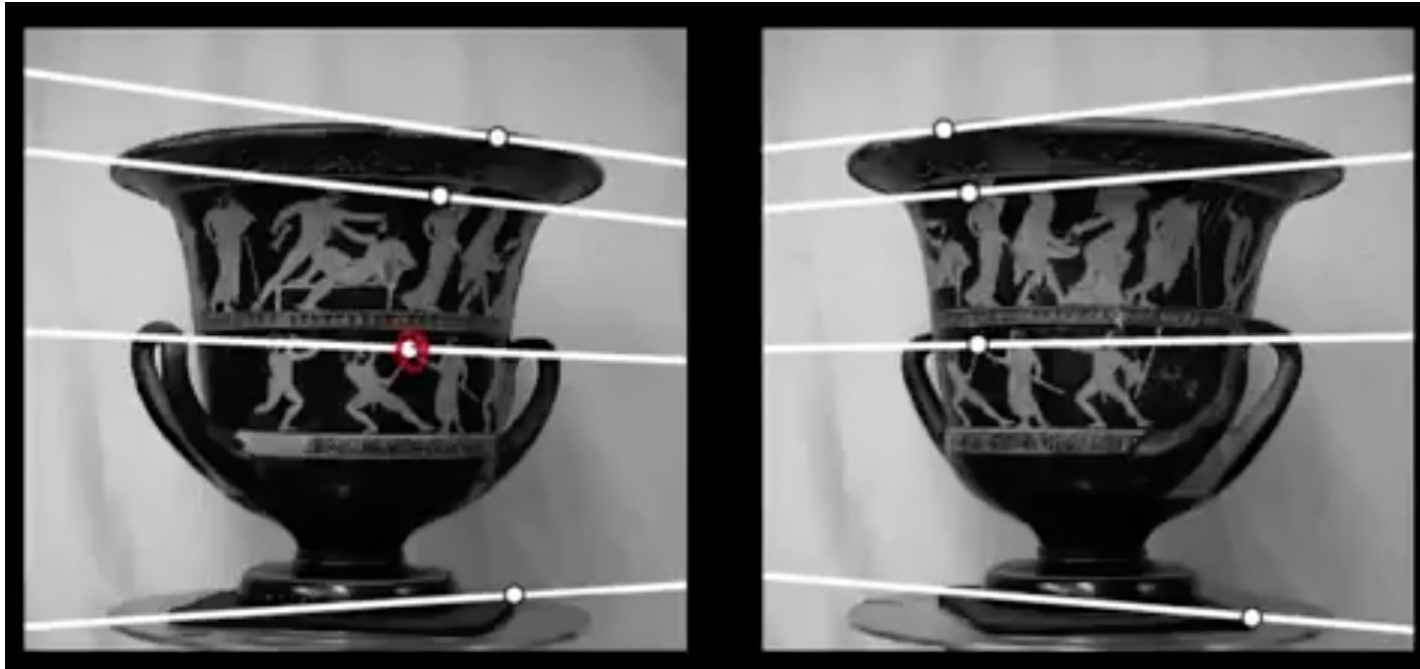
- We can search for matches across epipolar lines.
 - Search space for correspondences reduces to a 1D problem!

Epipolar Geometry



- We can search for matches across epipolar lines
 - Search space for correspondences reduces to a 1D problem!
- All epipolar lines intersect at the epipoles

Epipolar Geometry



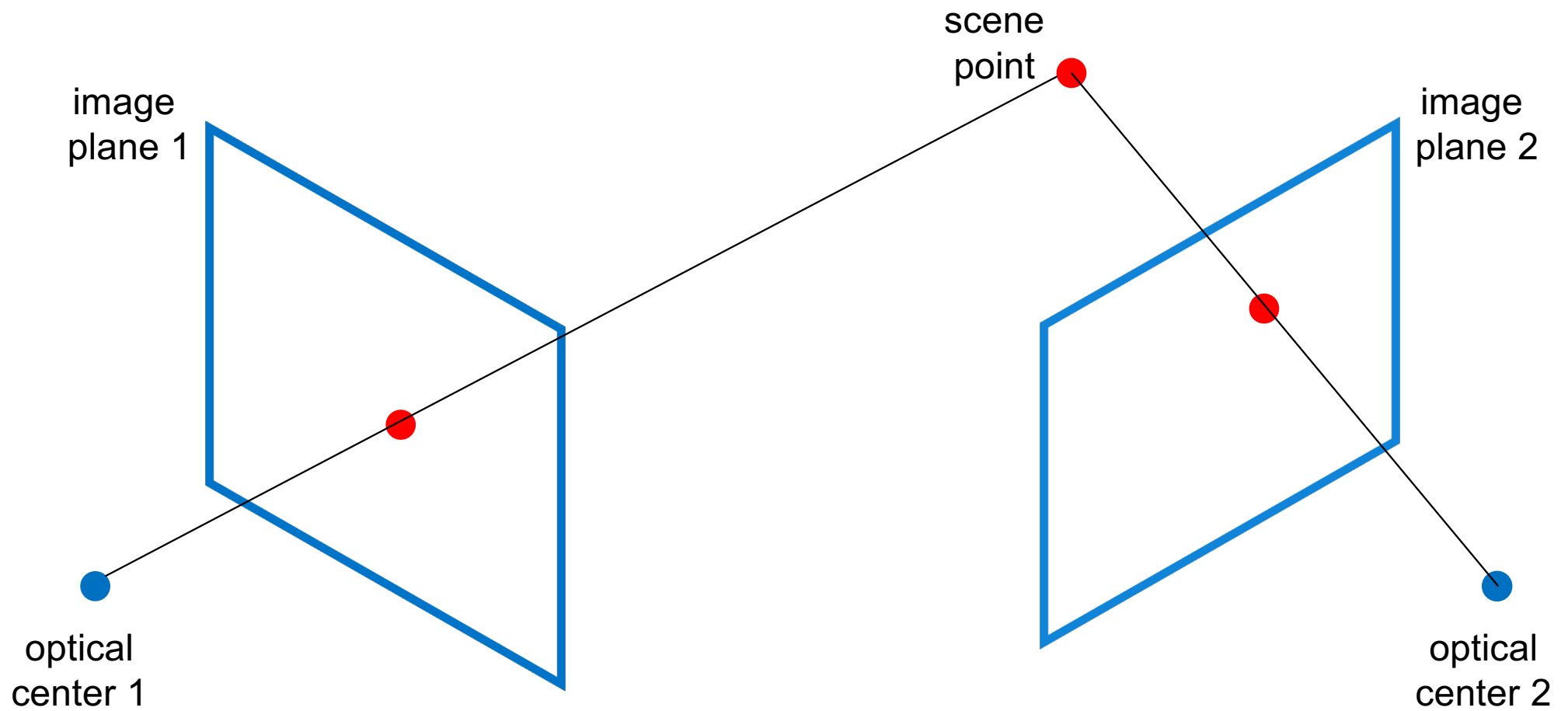
- We can search for matches across epipolar lines
 - Search space for correspondences reduces to a 1D problem!
- All epipolar lines intersect at the epipoles
 - **Where are the epipoles in this case?**



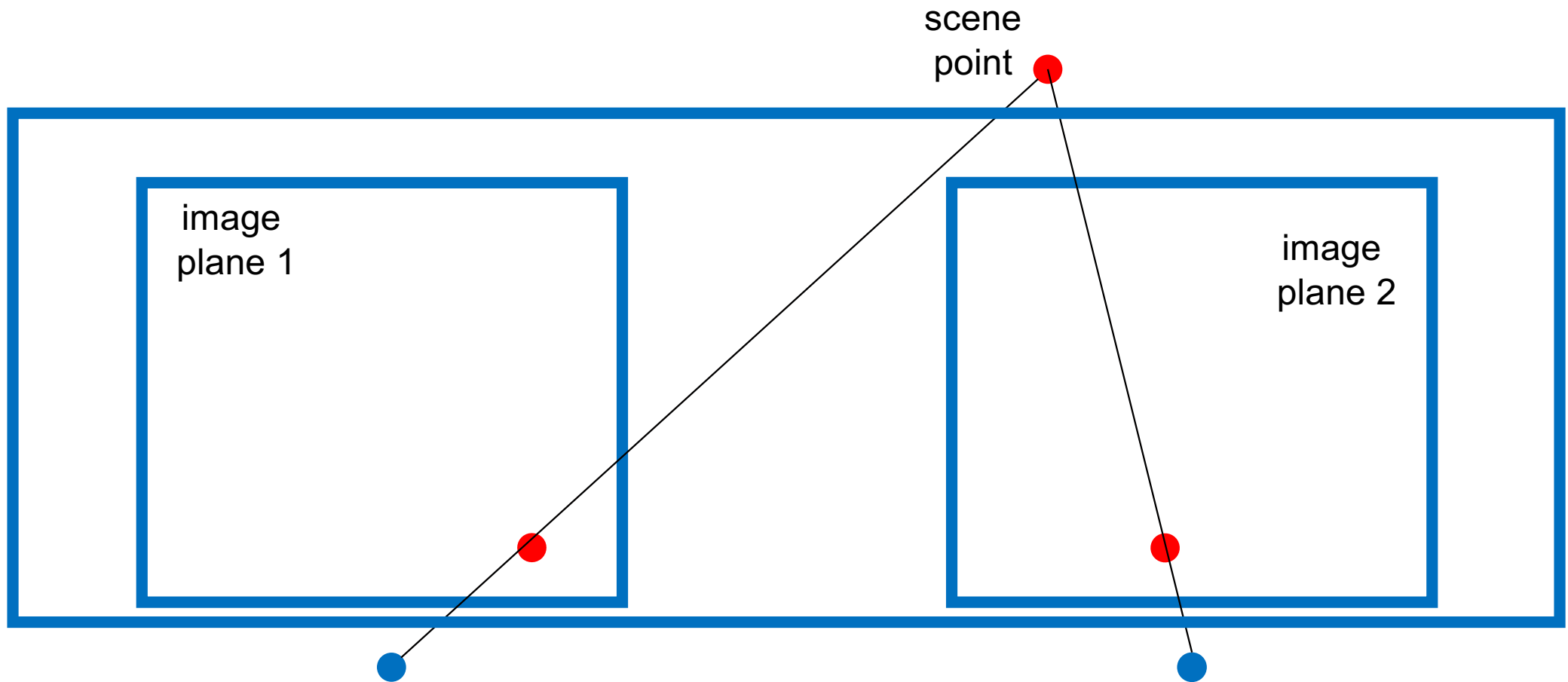
Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary

Rectified Stereo – a simpler case

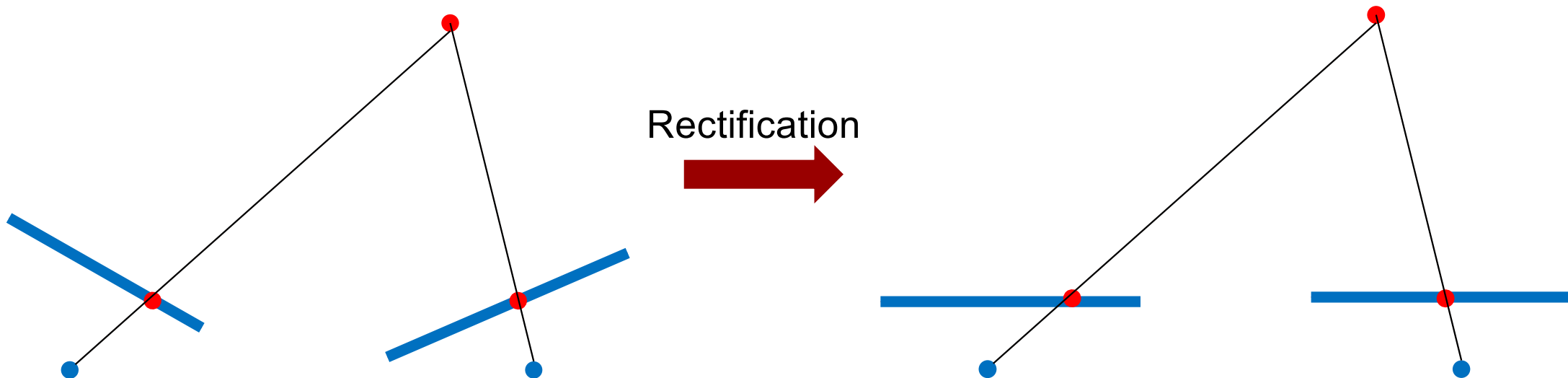


Rectified Stereo – a simpler case

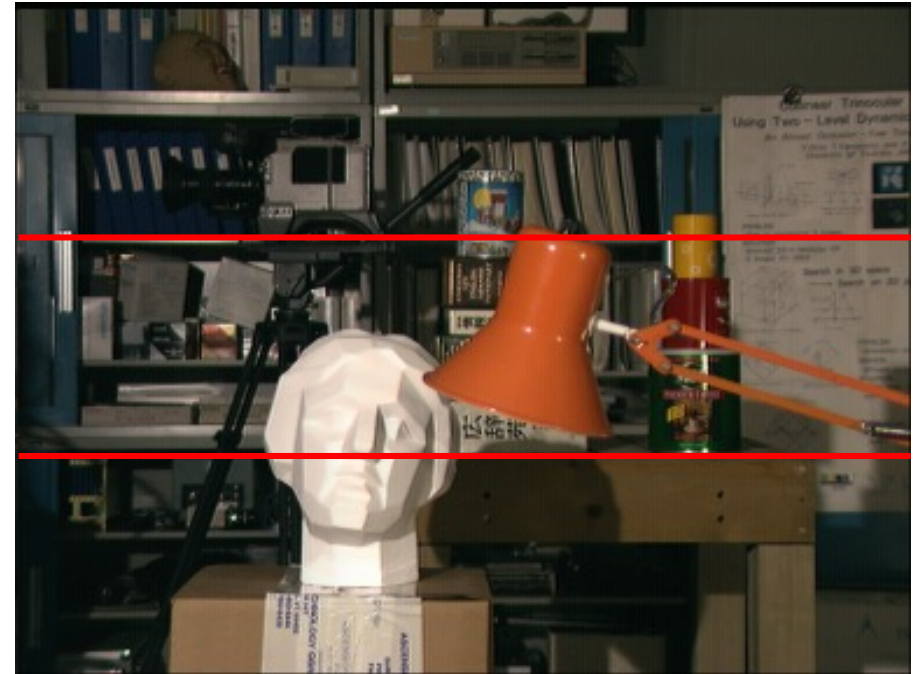
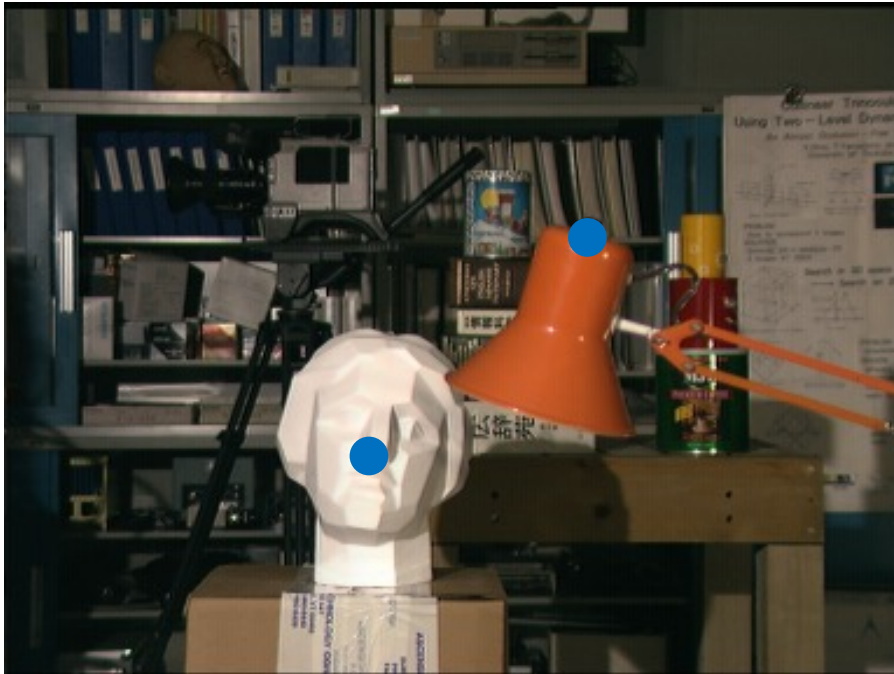


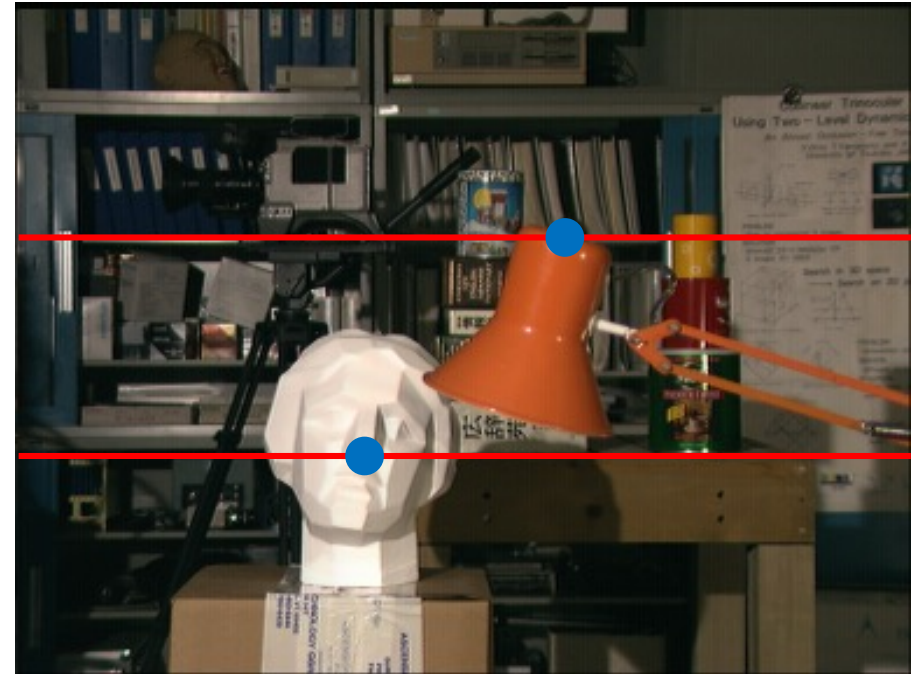
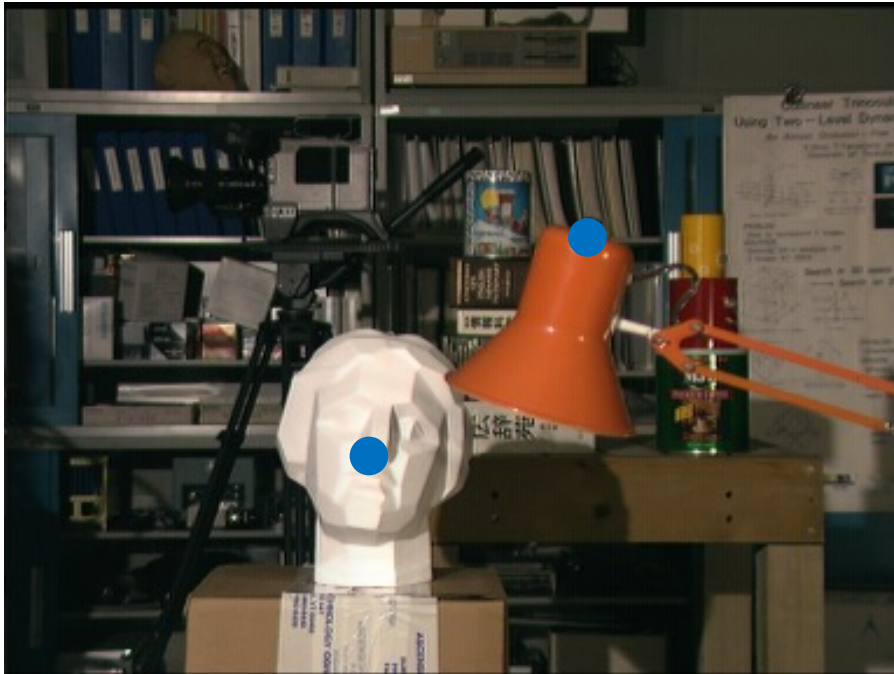
Rectified Stereo – a simpler case

- Rectification:
 - The initial images are reprojected on a common plane that is parallel to the baseline B joining the optical centers of the initial images.
 - Epipolar lines become parallel (and under certain conditions they become also horizontal)



DTU Rectified Stereo





- “All epipolar lines intersect at the epipoles”
 - **Where are the epipoles in this case?**



Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary

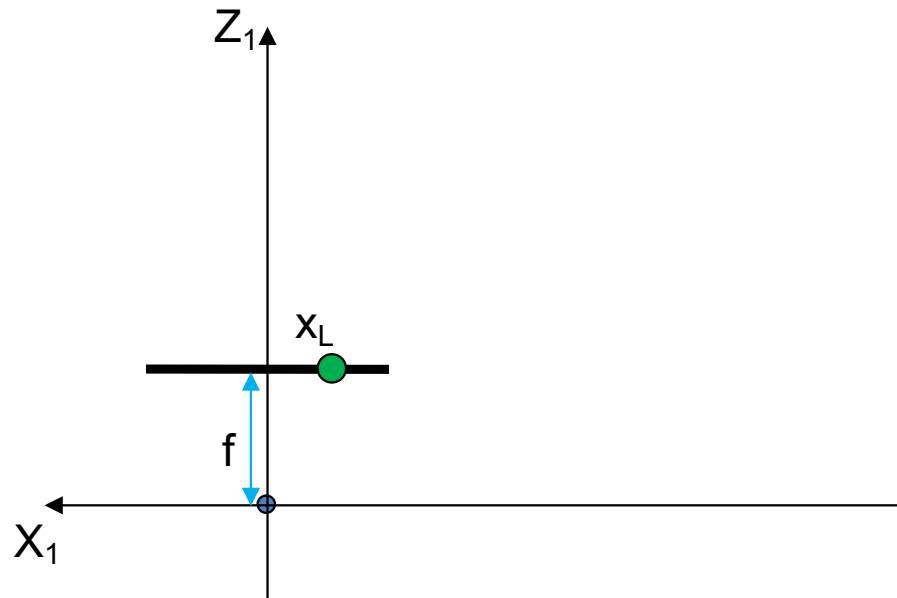


Depth from Stereo Matches

- Let us assume (for now!!) that:
 - we can check the points along the epipolar line and
 - we can find the point (on the right image) that is most similar to our reference point (in the left image), i.e. we can solve the correspondence problem!

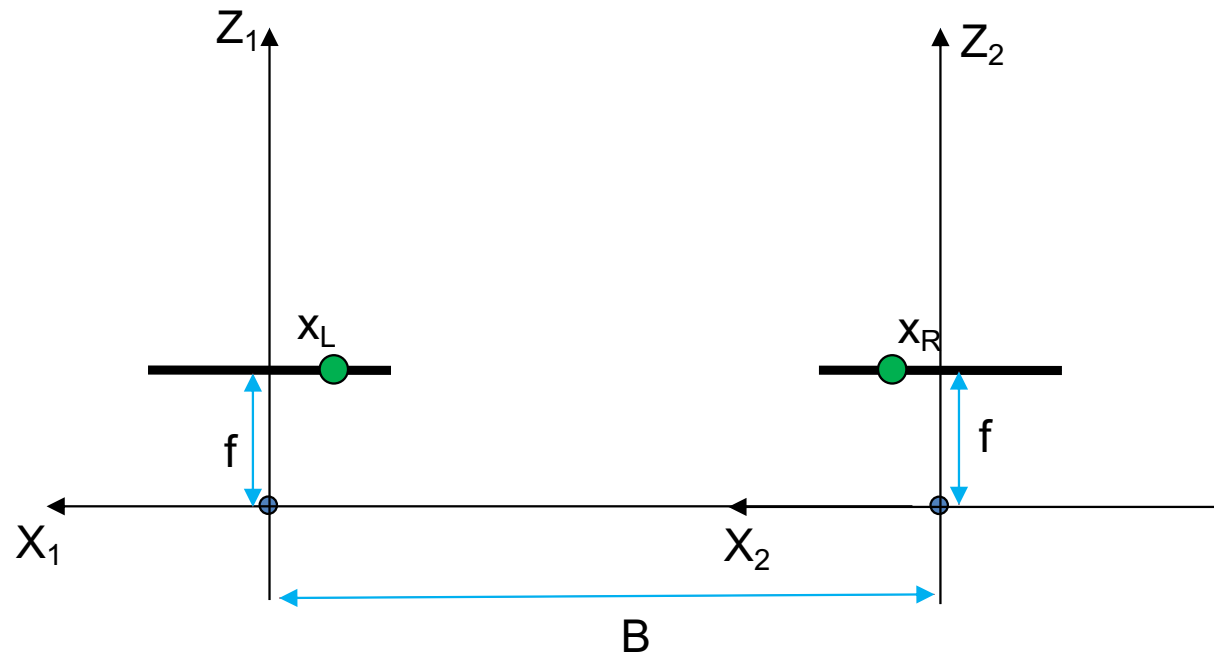


Depth from Stereo Matches



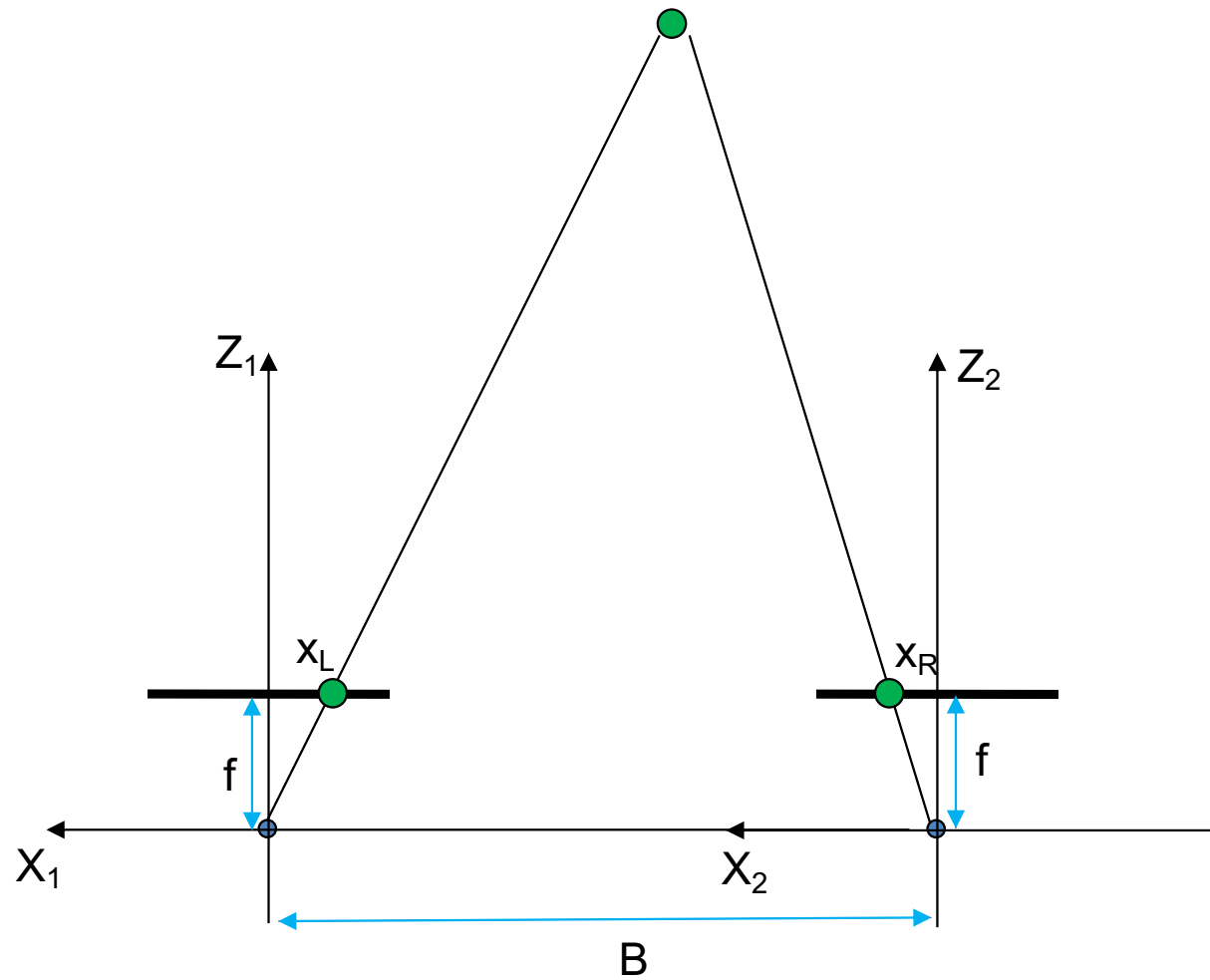


Depth from Stereo Matches



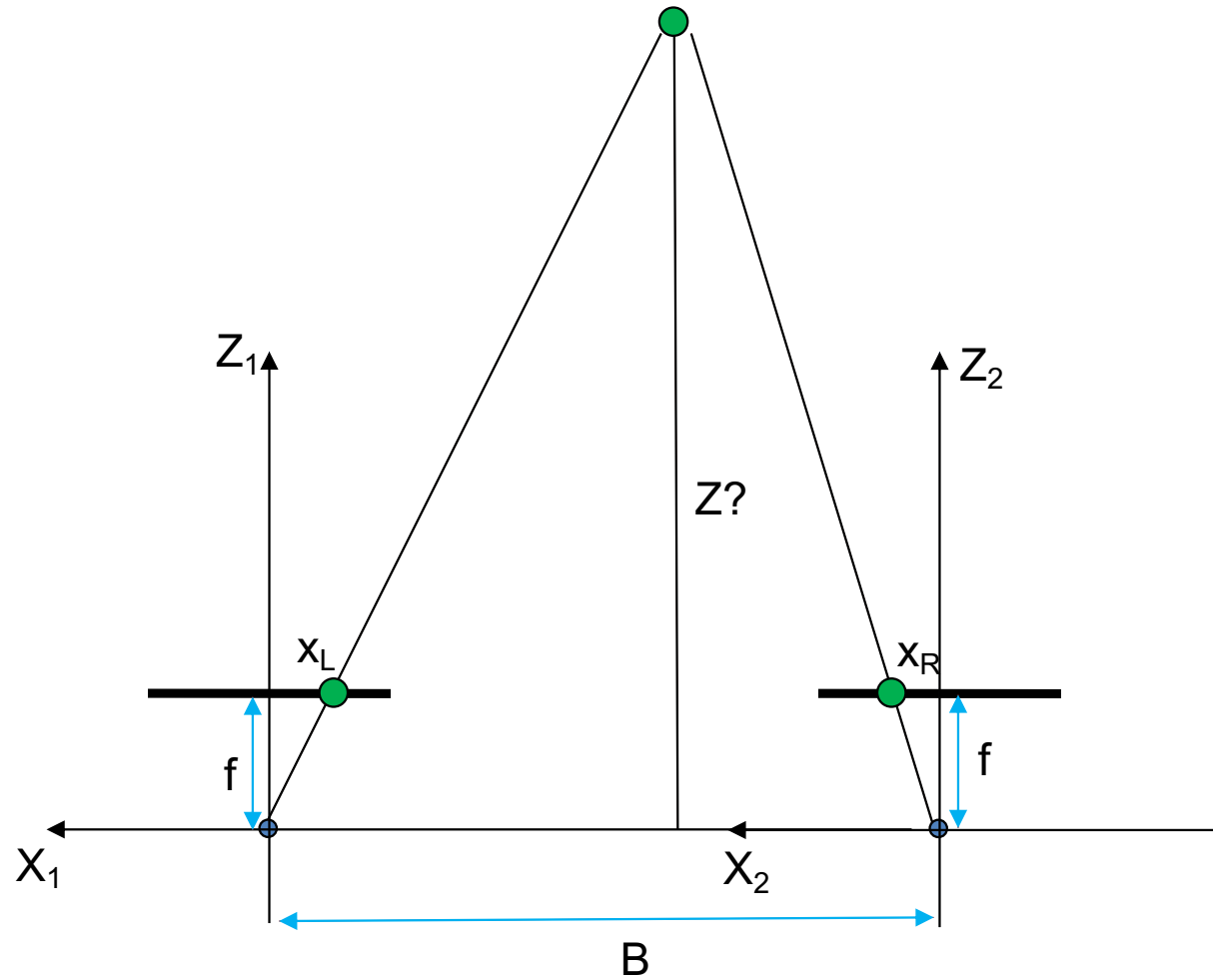


Depth from Stereo Matches



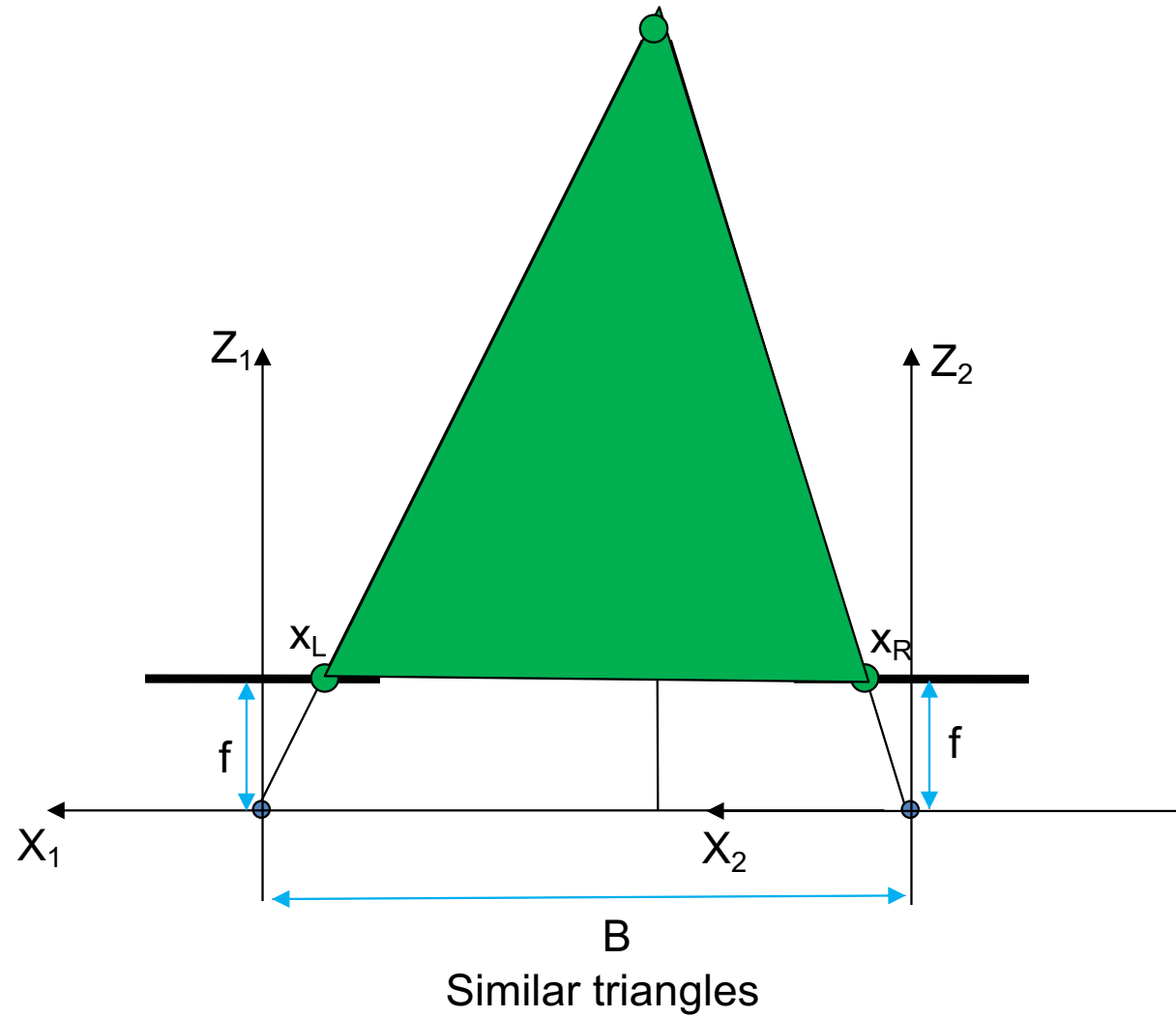


Depth from Stereo Matches



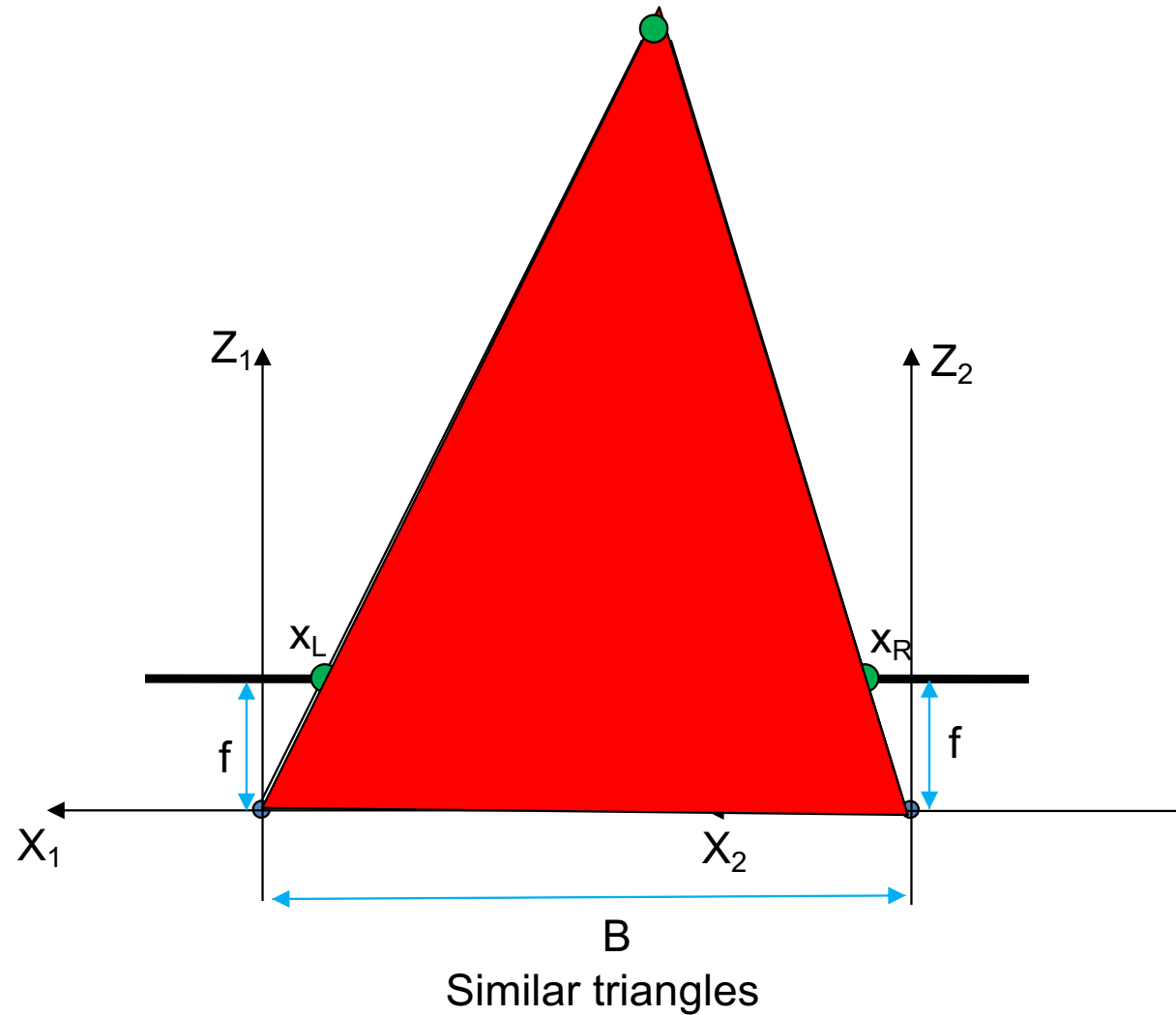


Depth from Stereo Matches

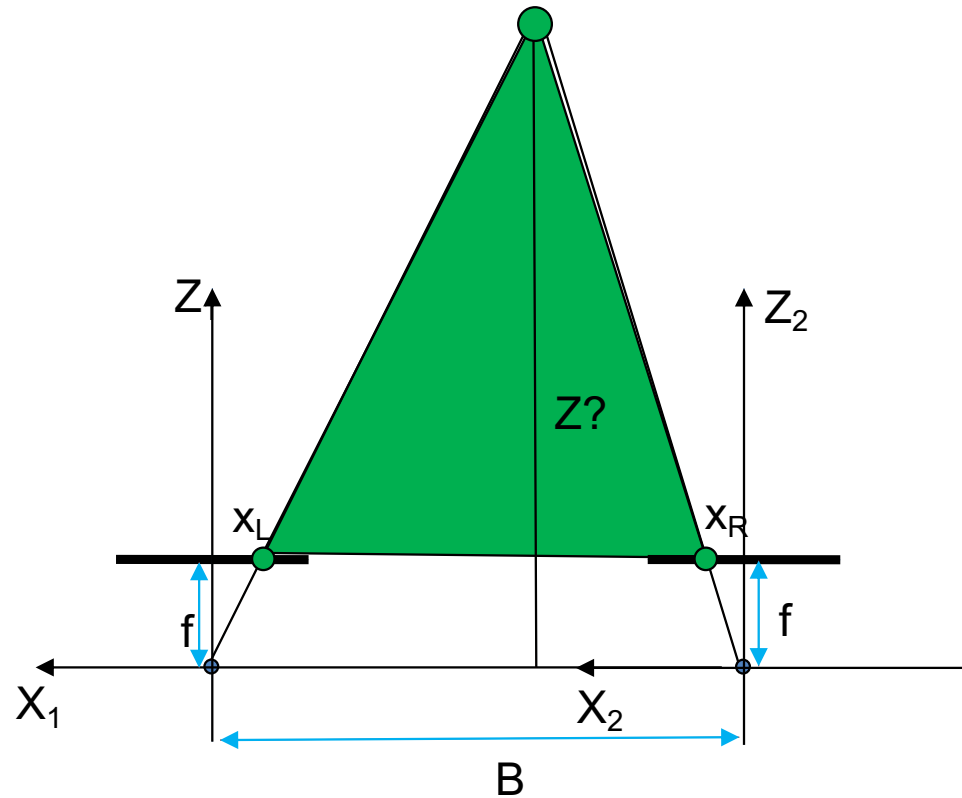




Depth from Stereo Matches

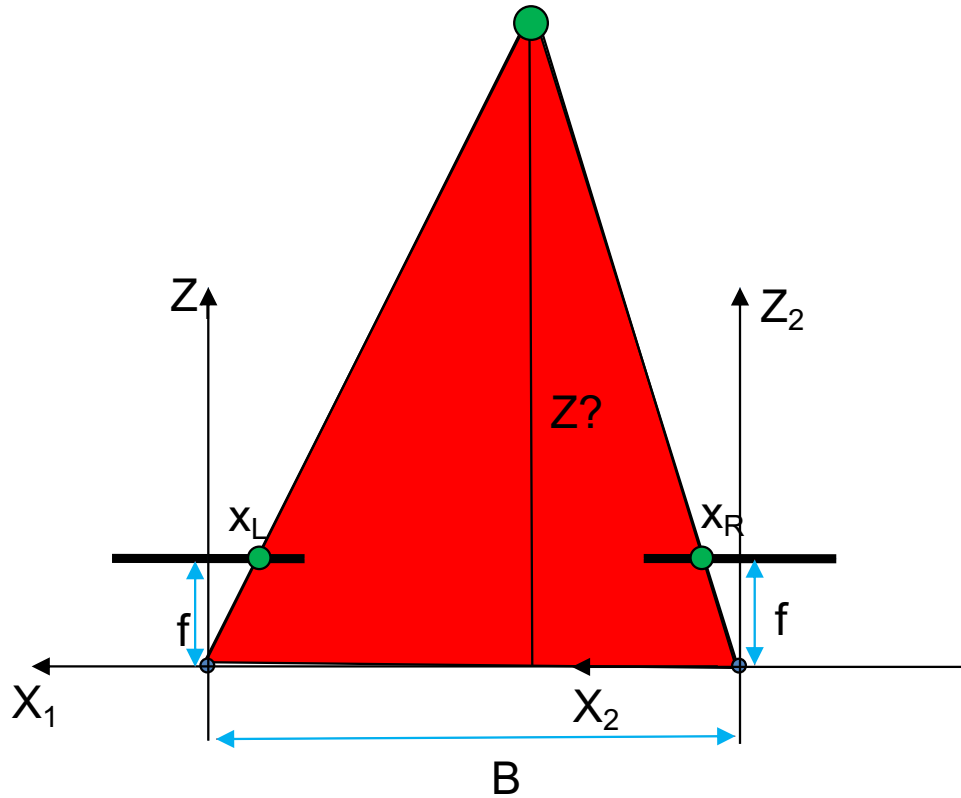


Depth from Stereo Matches



Similar triangles:

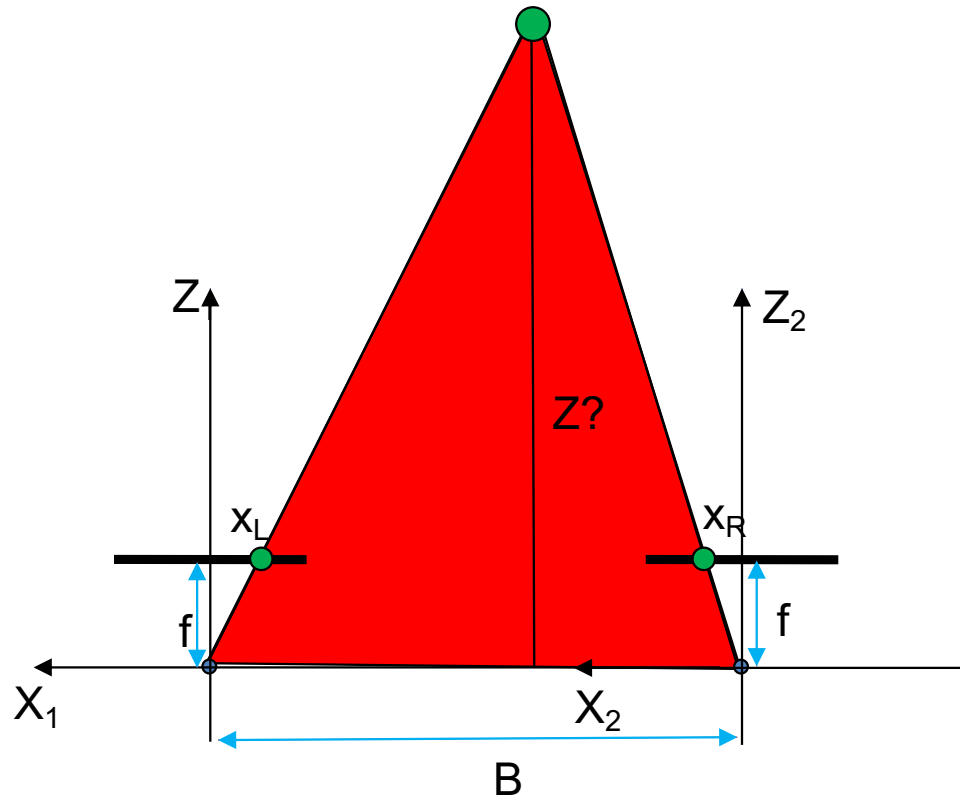
$$\frac{B - X_L + X_R}{Z - f} =$$



Similar triangles:

$$\frac{B - X_L + X_R}{Z - f} = \frac{B}{Z}$$

Depth from Stereo Matches



Similar triangles:

$$\frac{B - X_L + X_R}{Z - f} = \frac{B}{Z}$$

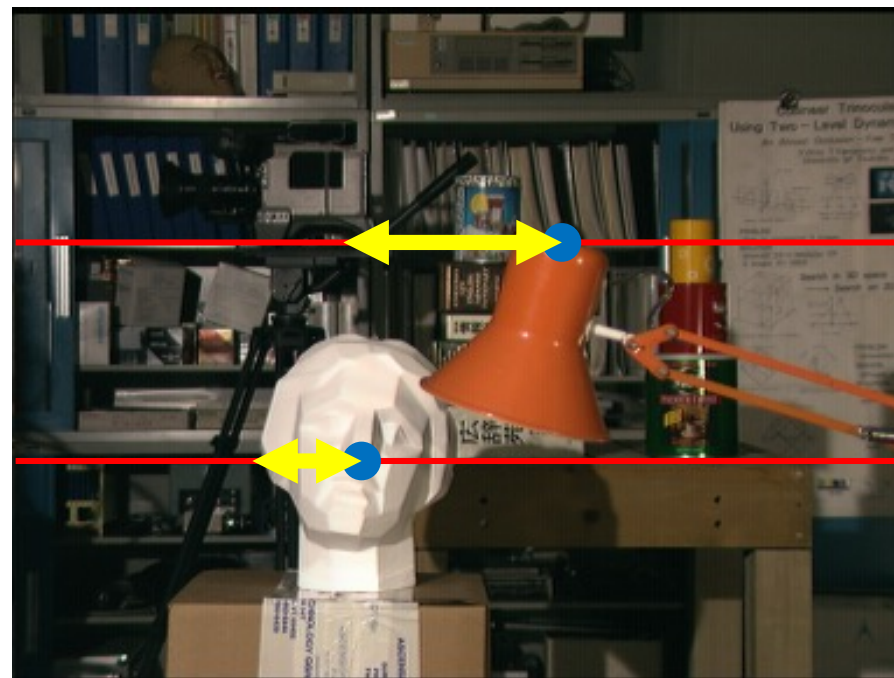
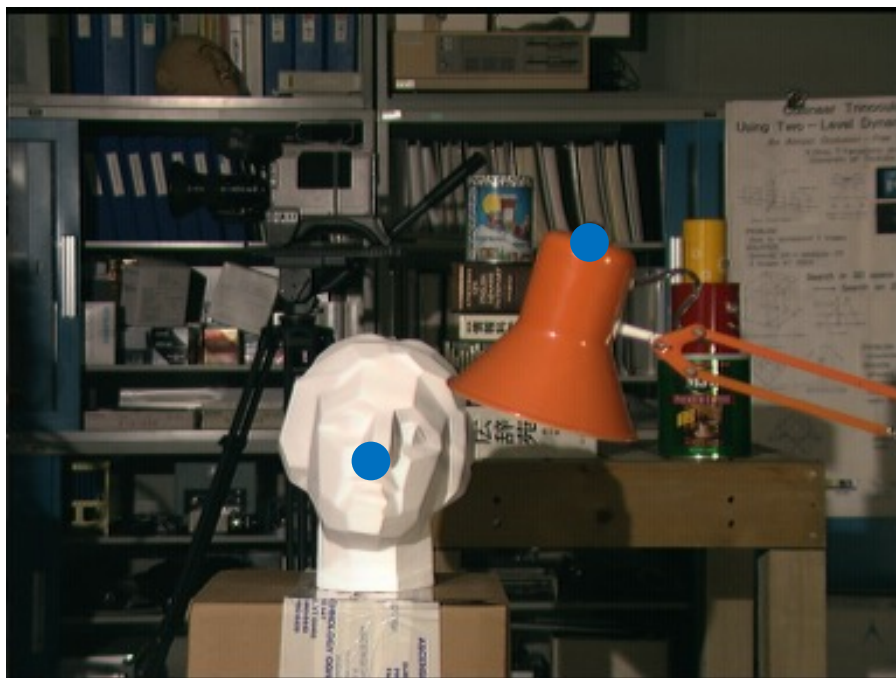
Solving for Z :

$$Z = f \frac{B}{X_L - X_R}$$

Disparity



Depth from Stereo Matches



Depth from Stereo Matches

Reference Image



Disparity Map



$$Z = f \frac{B}{X_L - X_R}$$

Disparity



Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary



Correspondence Problem

- We have assumed (up to now!!) that:
 - we can check the points along the epipolar line and
 - we can find the point (on the right image) that is **most similar** to our reference point (in the left image), i.e. we can **solve the correspondence problem!**
- *How can we indeed match corresponding pixels between the two stereo images?*



Correspondence Problem

- Beyond the hard constraint of epipolar geometry, there are “soft” constraints to help identify corresponding points
 - Similarity
 - Uniqueness
 - Ordering
 - Disparity gradient is limited



Correspondence Problem

- To find matches in the image pair, we will assume
 - Most scene points visible from both views
 - Image regions for the matches are similar in appearance



Correspondence Problem

- It depends!
 - Do we need **dense** or **sparse** stereo matching?

Stereo Vision

```
graph TD; A[Stereo Vision] --- B[Sparse output]; A --- C[Dense output]; C --- D[Local Methods (Area-based)]; C --- E[Global Methods (Energy-based)]; C --- F[Other Methods];
```

Sparse output

Dense output

- Local Methods (Area-based)
- Global Methods (Energy-based)
- Other Methods

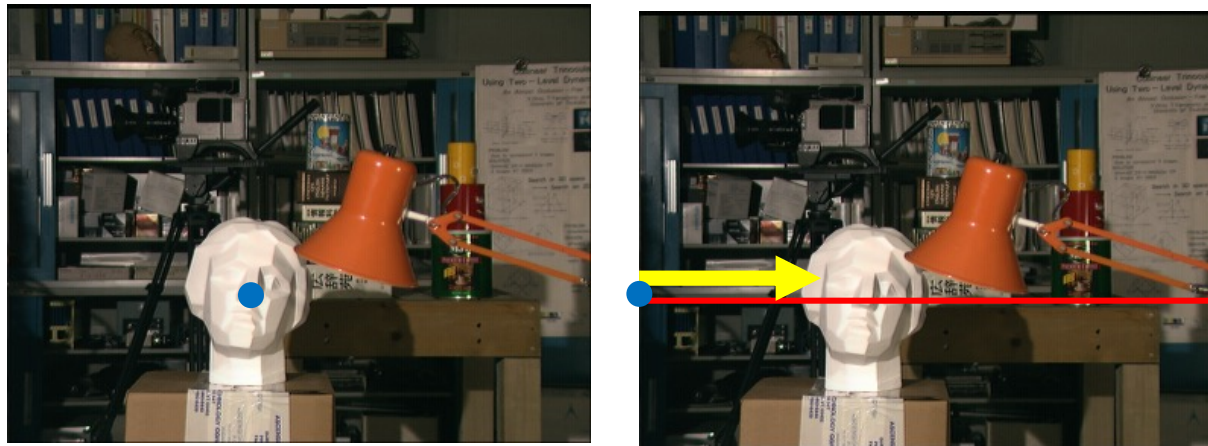


Sparse Stereo Correspondence

- Extract features (e.g. SIFT, SURF, Harris,...) and match them!
 - Pros?
 - Cons?

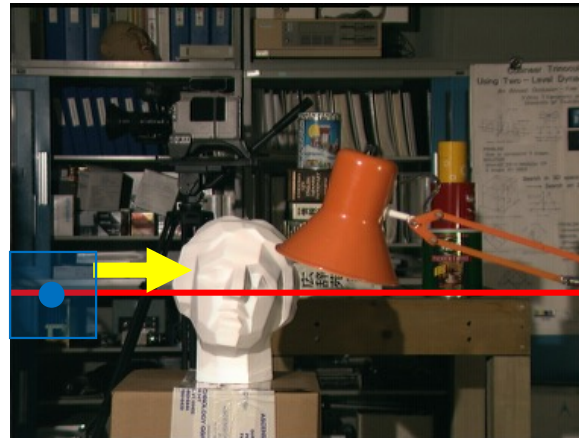
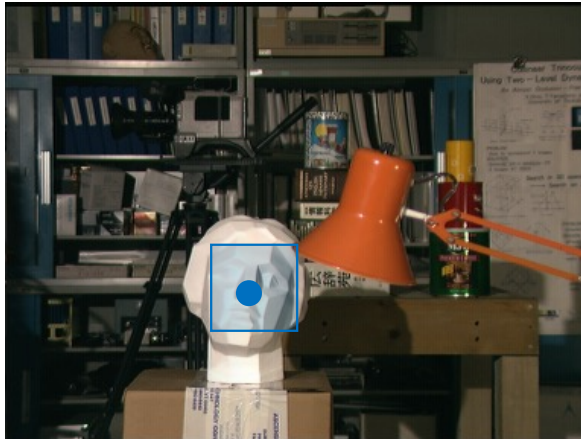
Dense Stereo Correspondence : Local Methods

- Try to find correspondences for all the pixels of the reference image.
- For each epipolar line
 - For each pixel in the left image
 - Compare with every pixel on same epipolar line in right image
 - Choose the pixel that maximizes a similarity metric (or minimizes a dissimilarity metric!).



Dense Stereo Correspondence

- Try to find correspondences for all the pixels of the reference image.
- For each epipolar line
 - For each pixel in the left image
 - Compare with every pixel on same epipolar line in right image
 - Choose the pixel that maximizes a similarity metric (or minimizes a dissimilarity metric!).
- Improvement: don't match individual pixels, but rather match windows!



Stereo Correspondence Metrics

- Sum of Absolute Differences (SAD)

$$SAD(x, y, d) = \sum_{x, y \in W} |I_l(x, y) - I_r(x, y - d)|$$

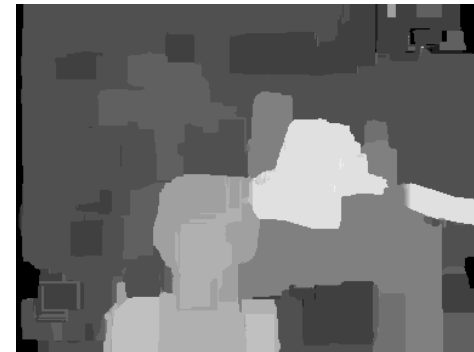
- Sum of Squared Differences (SSD)

$$SSD(x, y, d) = \sum_{x, y \in W} (I_l(x, y) - I_r(x, y - d))^2$$

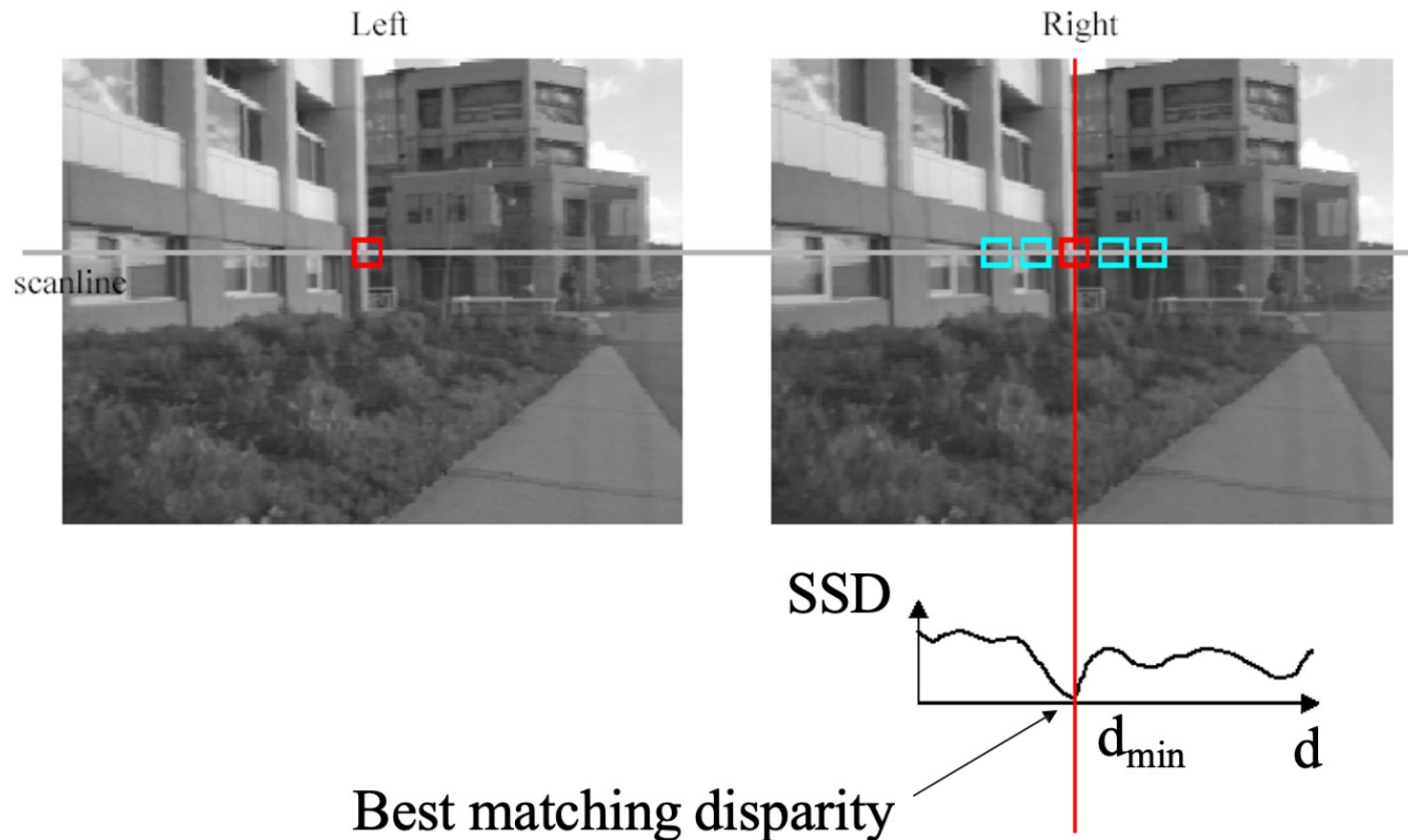
- Normalized Cross-Correlation

$$NCC(x, y, d) = \frac{\sum_{x, y \in W} I_l(x, y) \cdot I_r(x, y - d)}{\sqrt{\sum_{x, y \in W} I_l^2(x, y) \cdot \sum_{x, y \in W} I_r^2(x, y - d)}}$$

- ...many many more!!!



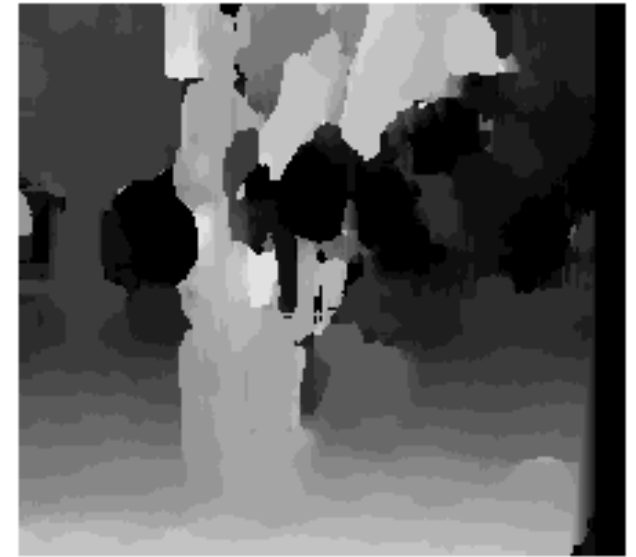
Stereo Correspondence Metrics: SSD



Stereo Correspondence Metrics: SSD on various windows



$W = 3$



$W = 20$

- Small vs Big windows
- What are their Pros and Cons?



Stereo Correspondence Metrics: Good/Bad areas



- In this stereo image pair:
 - what would be good areas to match?
 - where would you expect to face problems and why?



Global Stereo Correspondence

- Up to this point, the disparity of each pixel was determined only by the information of the pixel itself and its neighborhood.
 - Thus, those methods are called "local" or "area-based" methods.
- Example: Result of a **local** SSD algorithm with $W=21$:





Global Stereo Correspondence

- Up to this point, the disparity of each pixel was determined only by the information of the pixel itself and its neighborhood.
 - Thus, those methods are called "local" or "area-based" methods.
- Global methods find better solutions in expense of more computations
 - Optimize jointly the disparity values of all the pixels of each scanline (e.g. Dynamic Programming)



Global Stereo Correspondence

- Up to this point, the disparity of each pixel was determined only by the information of the pixel itself and its neighborhood.
 - Thus, those methods are called "local" or "area-based" methods.
- Global methods find better solutions in expense of more computations
 - Optimize jointly the disparity values of all the pixels of each scanline (e.g. Dynamic Programming)
 - Optimize jointly the disparity values of all the pixels of the image (e.g. graph cuts)





Global Stereo Correspondence

- Up to this point, the disparity of each pixel was determined only by the information of the pixel itself and its neighborhood.
 - Thus, those methods are called "local" or "area-based" methods.
- Global methods find better solutions in expense of more computations
 - Optimize jointly the disparity values of all the pixels of each scanline (e.g. Dynamic Programming)
 - Optimize jointly the disparity values of all the pixels of the image (e.g. graph cuts)
- *In global algorithms, stereo correspondence is formulated as an energy function minimization problem, consisting of data and smoothness terms.*

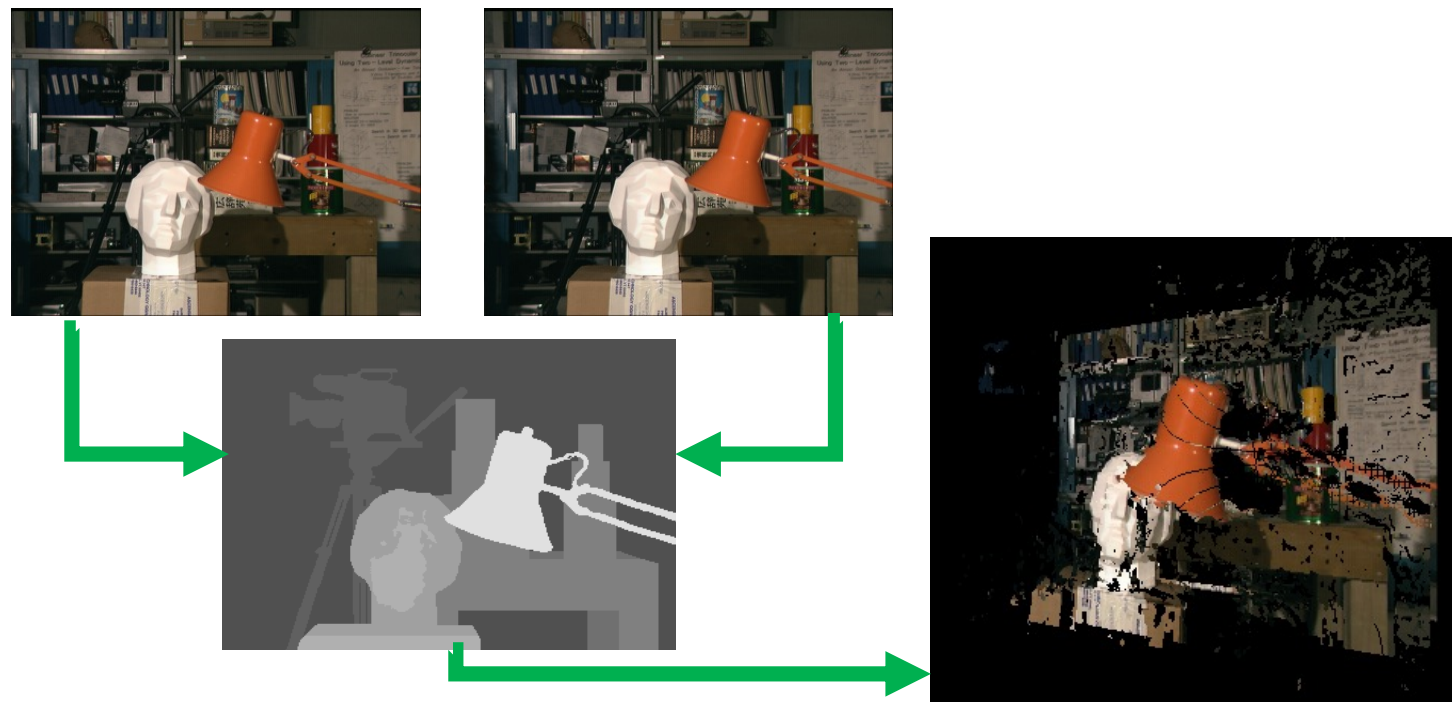


Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary

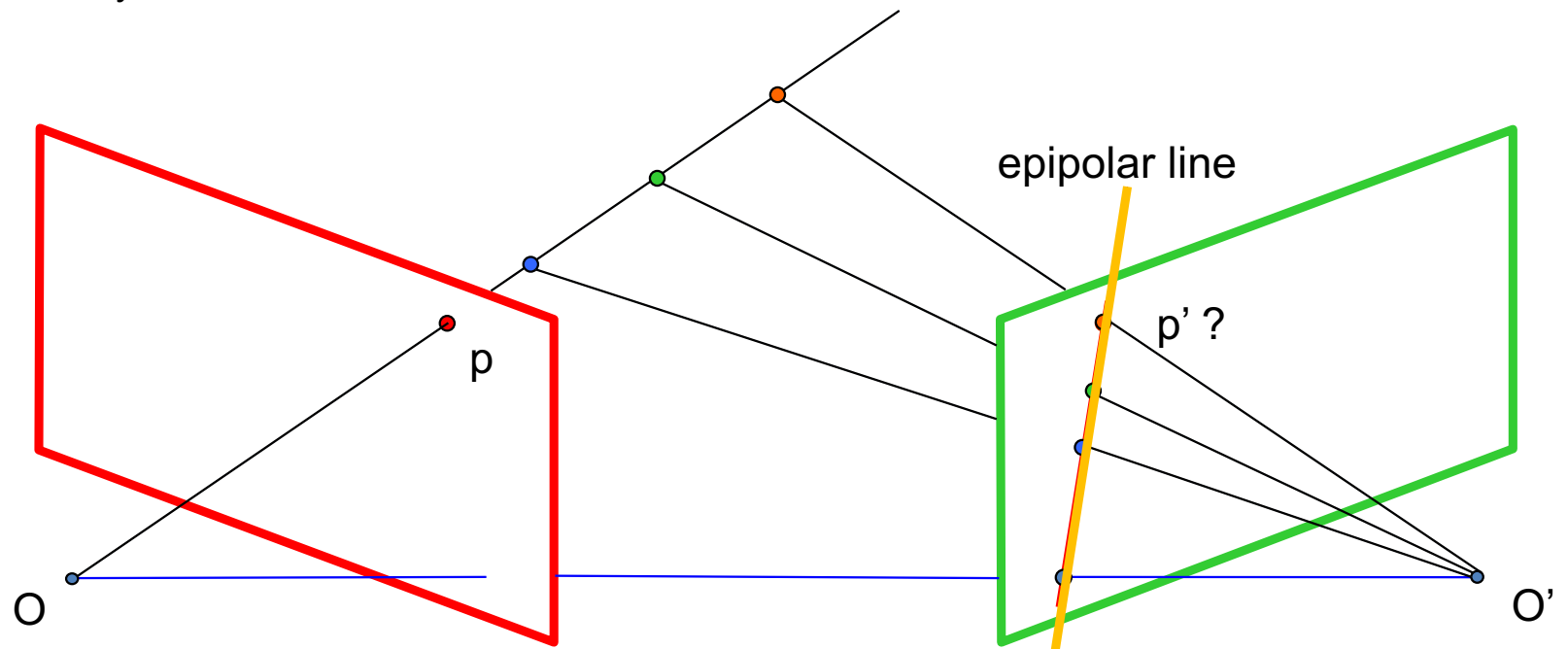
DTU Summary

- We discussed about what Stereo Vision is.



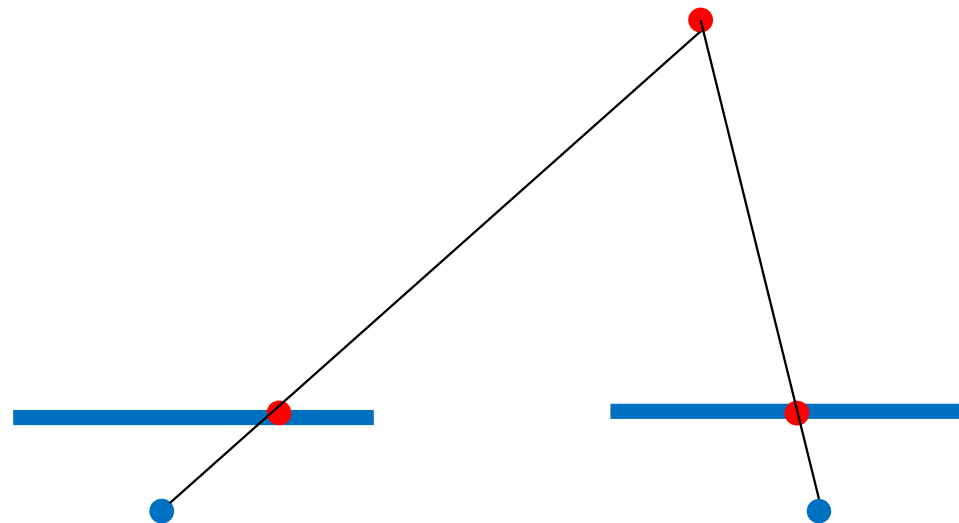
DTU Summary

- We discussed about what Stereo Vision is.
- We learned about :
 - Stereo/Epipolar Geometry



DTU Summary

- We discussed about what Stereo Vision is.
- We learned about :
 - Stereo/Epipolar Geometry
 - Rectified Stereo Case



DTU Summary

- We discussed about what Stereo Vision is.
- We learned about :
 - Stereo/Epipolar Geometry
 - Rectified Stereo Case
 - Depth from Stereo Matches

Reference Image



Disparity Map



$$Z = f \frac{B}{X_L - X_R}$$

Disparity



Summary

- We discussed about what Stereo Vision is.
- We learned about :
 - Stereo/Epipolar Geometry
 - Rectified Stereo Case
 - Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures

Stereo Vision

Sparse output

Dense output

- Local Methods (Area-based)
- Global Methods (Energy-based)
- Other Methods



Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary

One more thing...



Lightweight Monocular Depth Estimation through Guided Decoding

- M. Rudolph, Y. Dawoud, R. Güldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," *IEEE ICRA*, 2022.

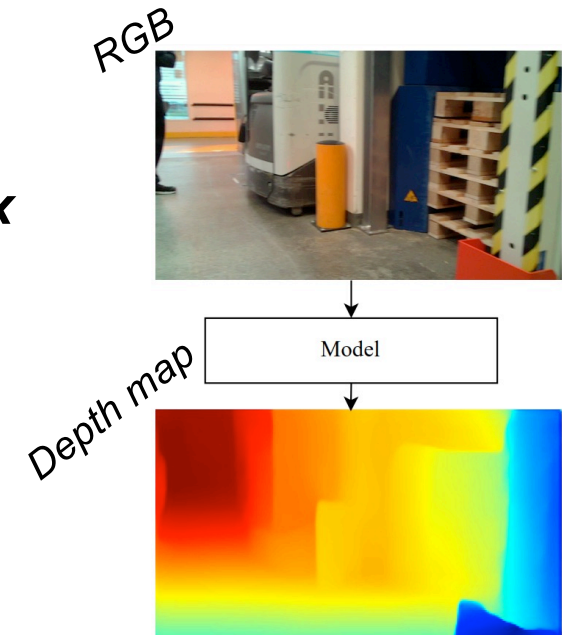
Lightweight Monocular Depth Estimation through Guided Decoding

Monocular Depth Estimation (MDE)

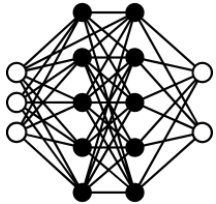
- Predict a **depth map y** from a **single input image x**
- **Inexpensive** compared to active sensors
- Recent advances thanks to **Deep Learning**

However:

- Large models with **many parameters**
→ Slow Inference
- We need **lightweight** and **accurate** models for robotic applications

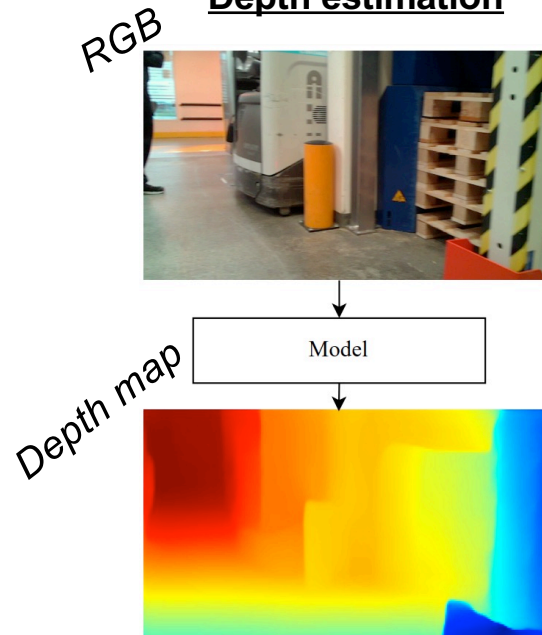


- M. Rudolph, Y. Dawoud, R. G ldenring, L. Nalpantidis, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," IEEE ICRA, 2022.



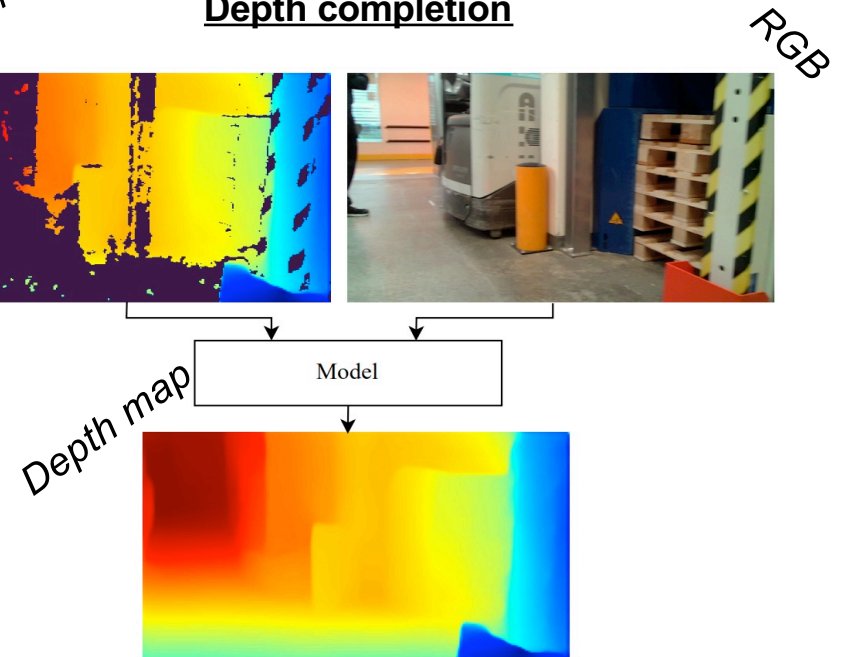
Use deep learning methods to infer depth of the scene and move from 2D to 3D.

Depth estimation



Sparse depth map

Depth completion



Lightweight Monocular Depth Estimation through Guided Decoding

Model operating on higher resolutions

- Capture details in the depth maps
- Balance accuracy and speed on NVIDIA Jetson devices

Easy deployment

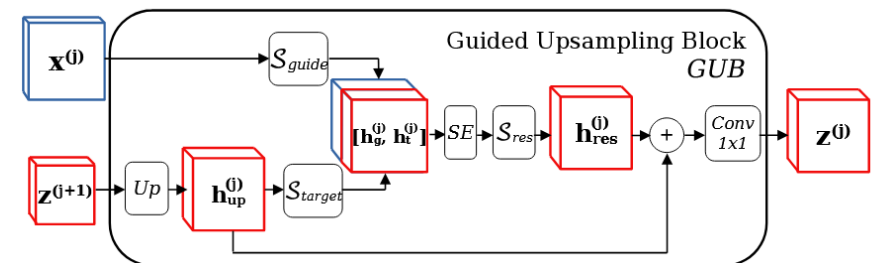
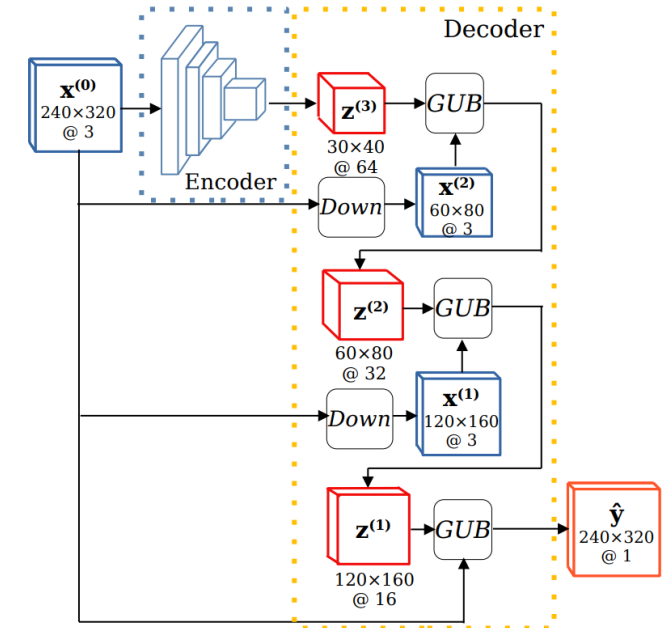
- Use tensorRT for inference

Use DDRNet as encoder

- Designed for real-time Semantic Segmentation

Design a Decoder based on feature guidance

- Use input image to guide the up-sampling process in the Decoder
- Allows to recover fine-grained details in the feature maps



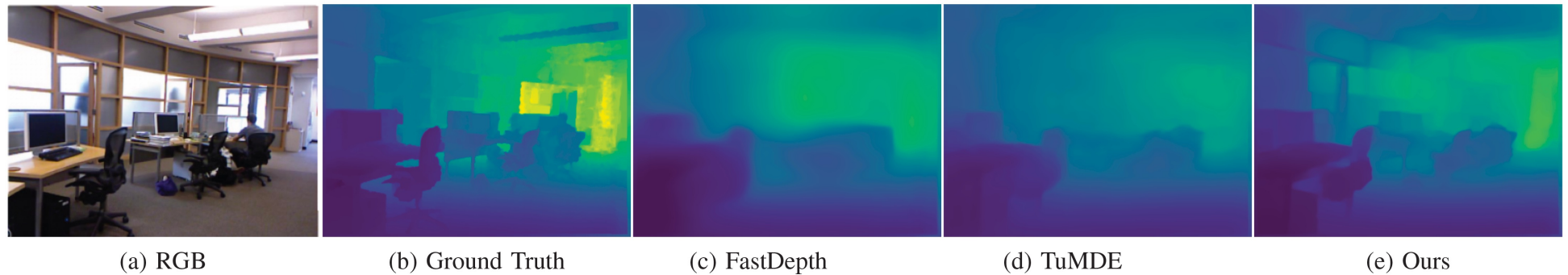
- M. Rudolph, Y. Dawoud, R. G ldenring, L. Nalpantidis, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," IEEE ICRA, 2022.

Lightweight Monocular Depth Estimation through Guided Decoding

- KITTI dataset:



- NYU v2 dataset:



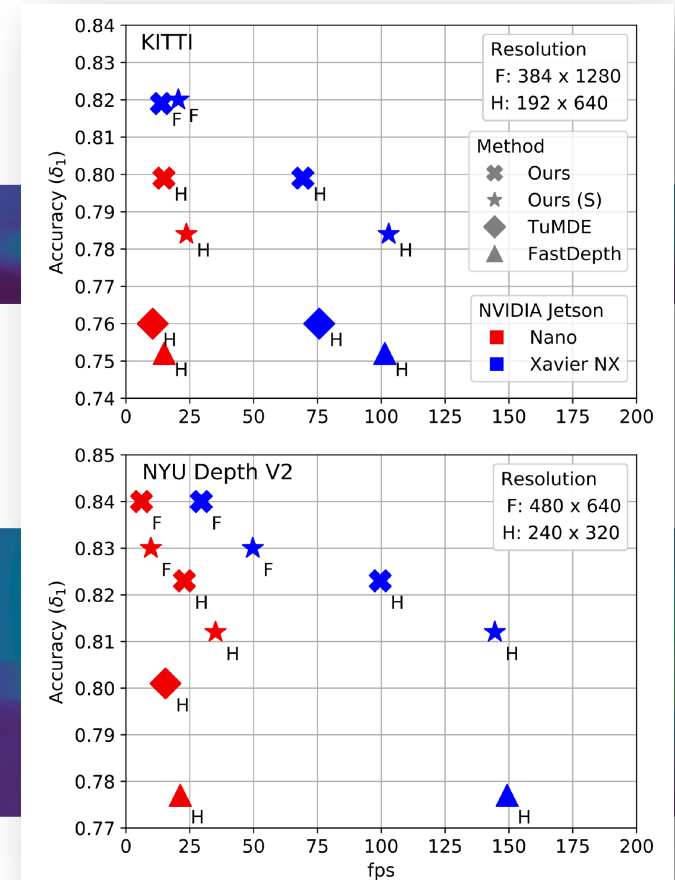
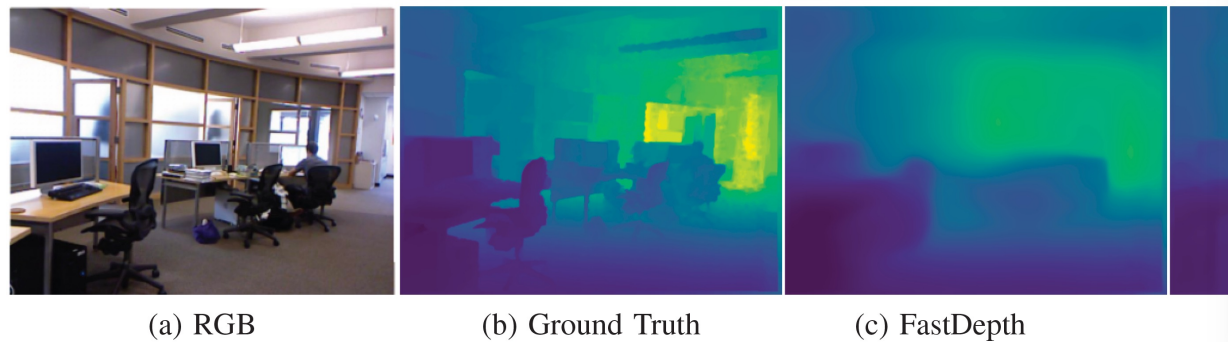
- M. Rudolph, Y. Dawoud, R. G ldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," *IEEE ICRA*, 2022.

Lightweight Monocular Depth Estimation through Guided Decoding

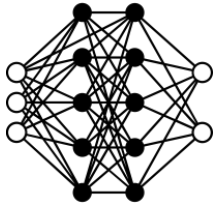
- KITTI dataset:



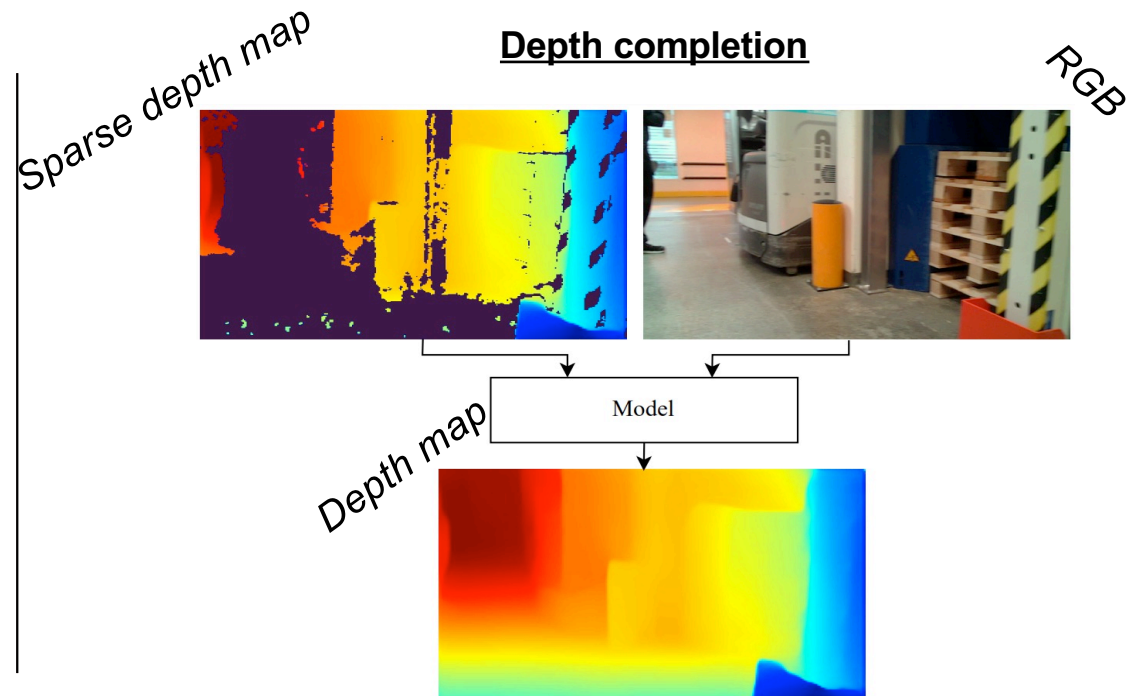
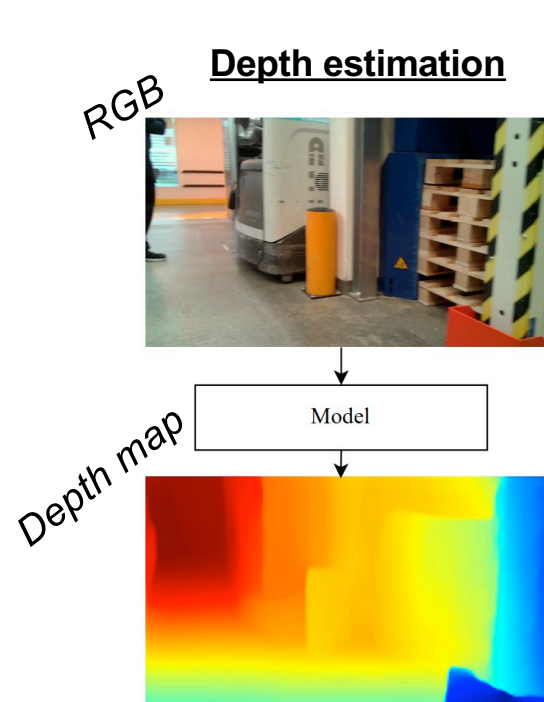
- NYU v2 dataset:



- M. Rudolph, Y. Dawoud, R. G ldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," IEEE ICRA, 2022.



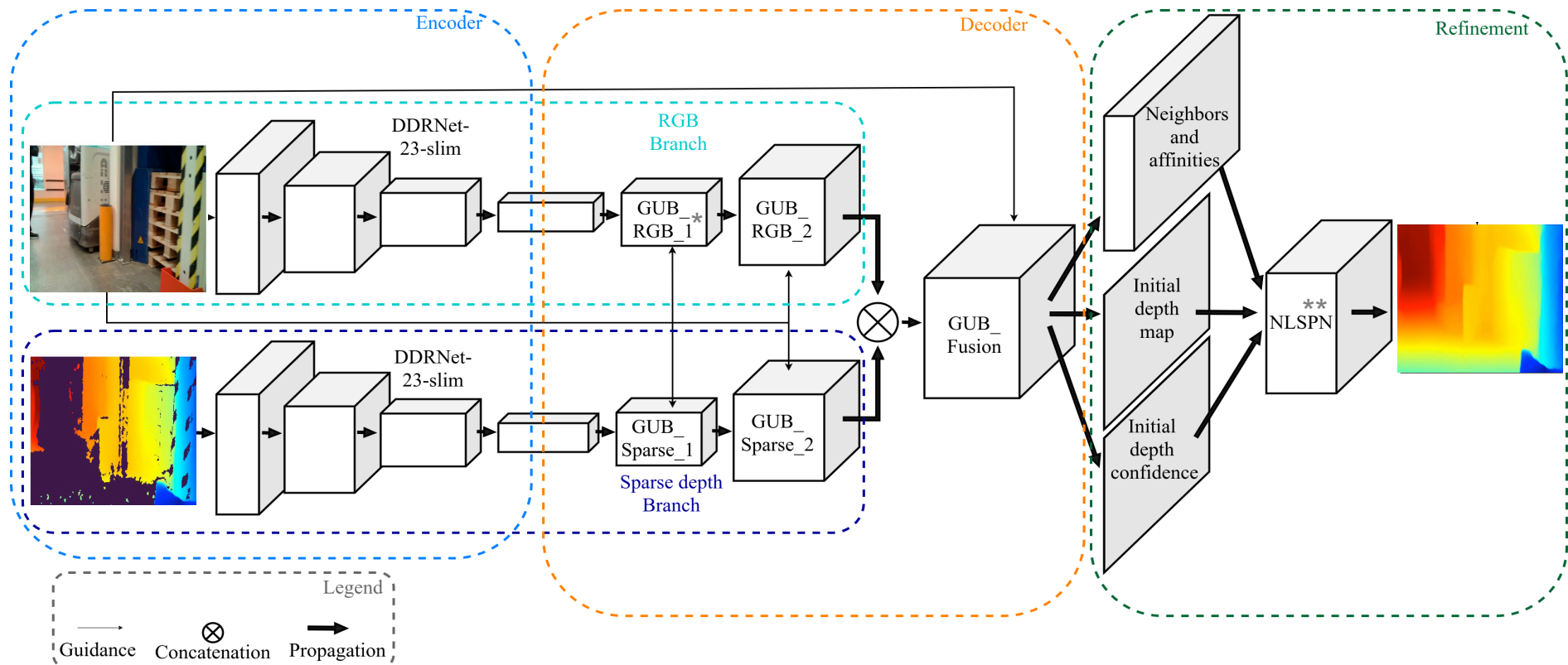
Use deep learning methods to infer depth of the scene and move from 2D to 3D.





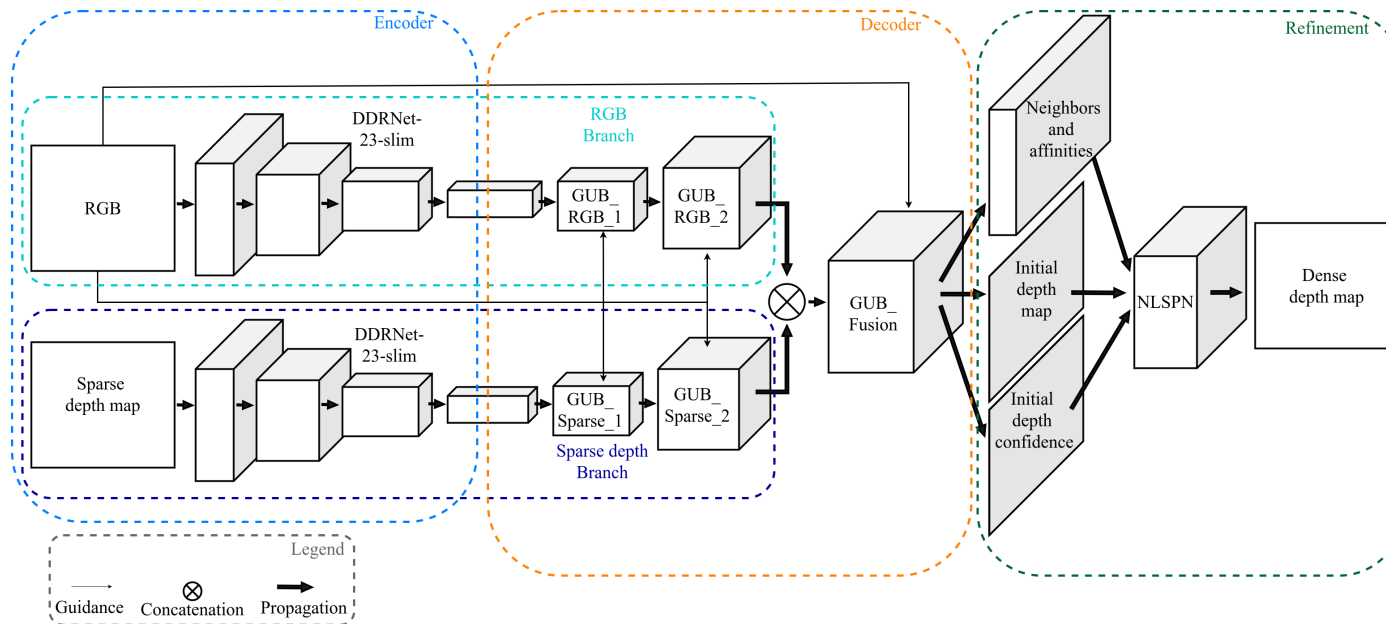
Goals:

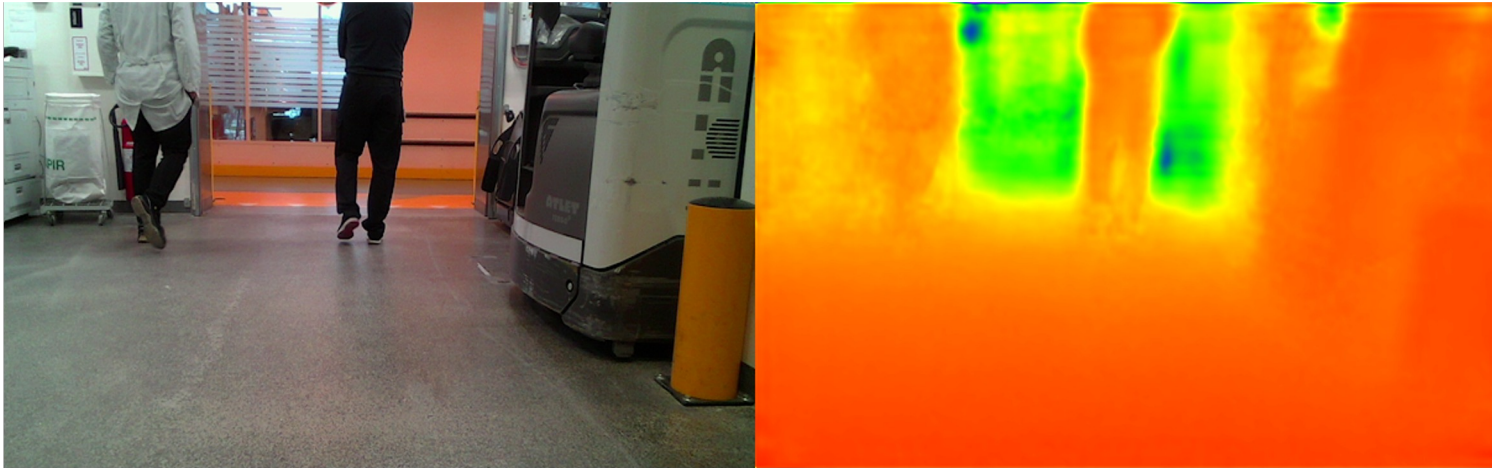
- Depth estimation model that can benefit from any available depth data → depth completion
- The model must be suitable for resource-constrained devices



- D. Arapis, M. Jami, and **L. Nalpantidis**, "Bridging depth estimation and completion for mobile robots reliable 3D perception," in *International Conference on Robot Intelligence Technology and Applications*, (Gold Coast, Australia), 2022.

Method	RMSE↓	REL↓	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	MACs(G)↓
(Our) LateFusion+NLSPN	0.099	0.014	99.4	99.9	99.9	10.7
LateFusion	0.116	0.018	99.2	99.8	99.9	7.1
EarlyFusion	0.114	0.018	99.3	99.8	99.9	10.0
EarlyFusion+NLSPN	<u>0.107</u>	<u>0.015</u>	<u>99.3</u>	<u>99.8</u>	<u>99.9</u>	13.6
Small	0.165	0.033	98.4	99.7	99.9	3.1
(Our_small) Small+NLSPN	0.111	0.017	99.3	99.8	99.9	4.0





Key findings



It is possible to **bridge depth estimation and completion** by designing networks that can incorporate **any amount of depth values to improve** their prediction.



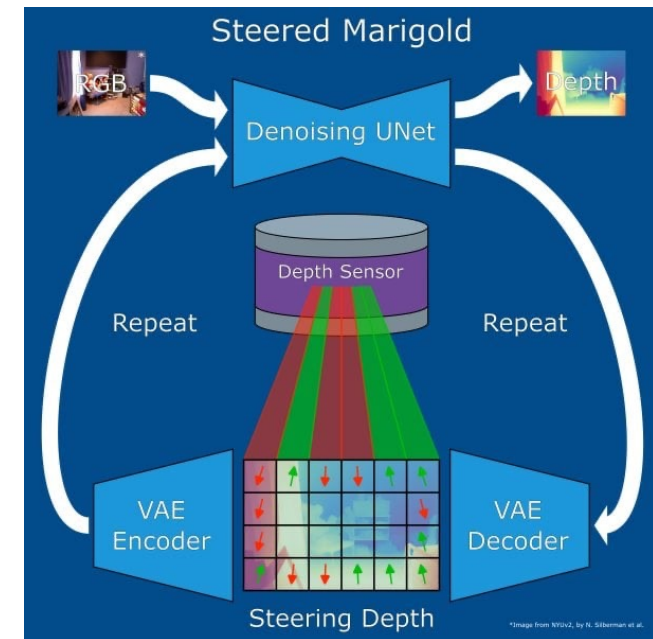
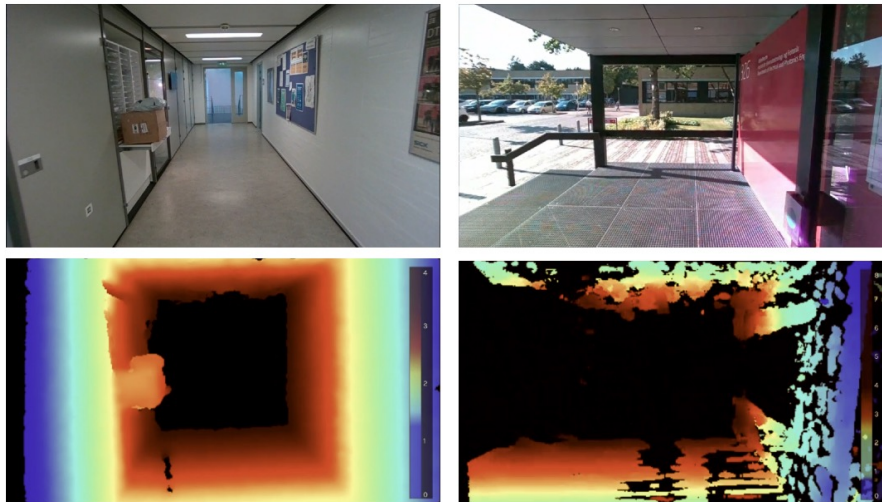
The proposed **refinement method** has a **bigger impact** in improving predictions compared to more complex backbones.



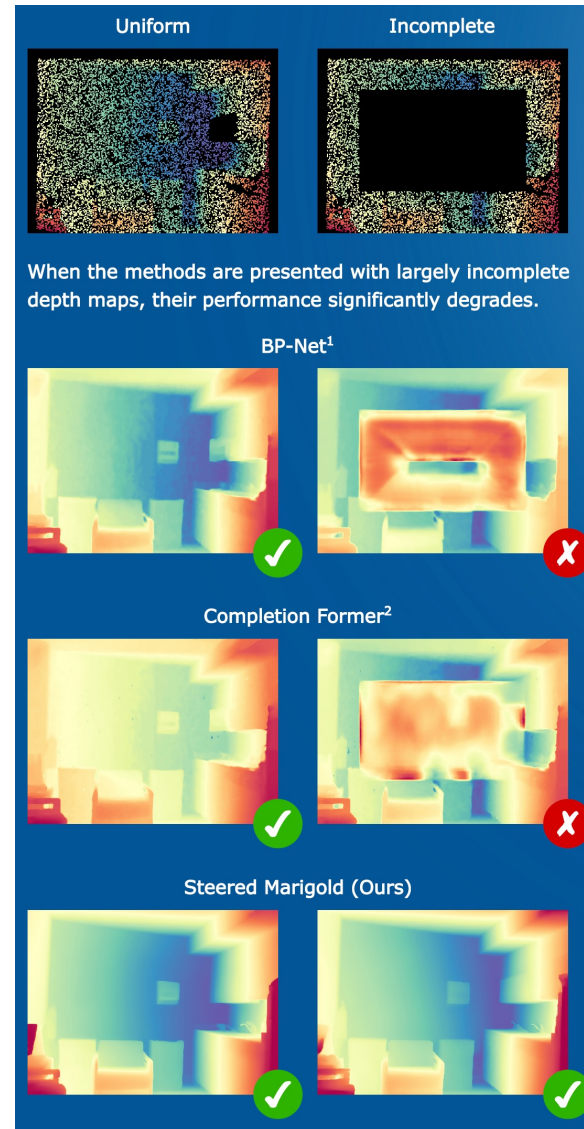
SteeredMarigold: Steering Diffusion Towards Depth Completion of Largely Incomplete Depth Maps

- *Jakub Gregorek and Lazaros Nalpantidis, "SteeredMarigold: Steering Diffusion Towards Depth Completion of Largely Incomplete Depth Maps," IEEE ICRA, 2025.*

SteeredMarigold



SteeredMarigold





Elite Summer School Robotics and Entrepreneurship
August 14, 2025
Odense, Denmark

Lazaros Nalpantidis

Perceiving Depth from Images

Stereoscopic Vision and the Stereo Correspondence Problem